Dyadic Weil Representations, Clifford Groups, and Theta Correspondence

Finite Quotients, Tensor Decompositions, and Local Theta Lifts

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Abstract

For a finite extension $F/\mathbb{Q}_2$, fix an additive character with conductor ideal $4\mathcal{O}_F$. The Weil representation contains a lattice model which factors through a nonsplit double cover of $\mathrm{Sp}_{2n}(\mathcal{O}_F/4\mathcal{O}_F)$. We determine its image, center, kernel, and restriction to the level-two congruence subgroup. Restriction of scalars and stabilization compare it with Paull's finite models. Over $\mathbb{Q}_2$, we decompose the fourth tensor power of the Clifford representation and construct explicit maps from its multiplicity spaces to local theta lifts for $O(\mathbb{Q}_2^4, I_4)$. We identify the lift of the trivial representation by Langlands data and determine the submodule lattice of the Siegel degenerate principal series at $-1/2$ over dyadic fields. The orthogonal translates of the lattice model yield eight irreducible orthogonal representations. Their first-occurrence indices and Jacquet modules determine standard-module expressions for the higher lifts. Seven tempered or supercuspidal inducing representations remain to be identified independently of their theta construction.

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1 Notation and conventions

We study finite quotients of dyadic Weil representations and their relation to Clifford groups and local theta correspondence. All representations are complex; representations of locally profinite groups are smooth. A dyadic field here is a finite extension of $\mathbb{Q}_2$, not a field of characteristic two.

The argument has four stages: finite descent and comparison with Paull (Sections 2–7); finite tensor decompositions and covering groups (Sections 8–11); the local theta lift of the trivial orthogonal representation and its induced realizations (Sections 12–14); and nontrivial orthogonal representations, their multiplicity spaces, and first occurrence (Sections 15–17). The appendices contain the lifting proof, notation, and exact finite computations used in the text.

1.1 The Pauli group and the full unitary Clifford group

On $\mathcal{H}_N = \mathbb{C}[\mathbb{F}_2^N]$, define

$$X_a e_x = e_{x+a}, \qquad Z_b e_x = (-1)^{b\cdot x} e_x, \qquad a, b, x \in \mathbb{F}_2^N.$$

The finite Pauli group is

$$P_N = \{i^s X_a Z_b : s \in \mathbb{Z}/4,\ a, b \in \mathbb{F}_2^N\}, \quad Z(P_N) = \mu_4 I, \quad |P_N| = 2^{2N+2},$$

with commutation relation

$$Z_b X_a = (-1)^{a\cdot b} X_a Z_b.$$

The full unitary Clifford group is

$$\mathrm{Cl}_N = N_{U(2^N)}(P_N).$$

It contains all scalars $U(1)I$, and is therefore not finite. Its finite projective quotient fits into

$$1 \longrightarrow \mathbb{F}_2^{2N} \longrightarrow \mathrm{Cl}_N / U(1) \longrightarrow \mathrm{Sp}_{2N}(\mathbb{F}_2) \longrightarrow 1. \tag{1.1}$$

1.2 Heisenberg groups with centers C_2 and C_4

Let $W = \mathbb{F}_2^{2N}$ carry a nondegenerate alternating form $\langle\cdot,\cdot\rangle$, choose a bilinear form $B : W \times W \to \mathbb{F}_2$ such that $B(v,w) + B(w,v) = \langle v, w\rangle$, and set

$$H(B) = W \times \mathbb{F}_2, \qquad (v,t)(w,s) = (v+w, t+s+B(v,w)).$$

The commutator is governed by $B(v,w) + B(w,v)$, whereas squaring also records

$$(v,t)^2 = (0, Q(v)), \qquad Q(v) = B(v,v).$$

An automorphism fixing the center must preserve this square relation. Its induced linear action on the label space therefore preserves Q and belongs to $O(Q) \subseteq \mathrm{Sp}(W)$. This is in general a

proper restriction, but equality holds for the anisotropic quadratic form in dimension two over $\mathbb{F}_2$. This explains the appearance of the pseudosymplectic group in Chapter 1 of Paull's thesis.

After adjoining fourth roots of unity, one may write the finite Heisenberg group as

$$H_4 = W \times \mathbb{Z}/4, \qquad ((x,y),s)((x',y'),s') = ((x+x',y+y'),s+s'+2x\cdot y').$$

The central character is $s \mapsto i^s$. Suitable $\mathbb{Z}/4$-valued central corrections compensate for the change of this cocycle under a symplectic transformation. This yields an affine symplectic group surjecting onto the whole of $\mathrm{Sp}(W)$. In the Pauli model it can be identified with $\mathrm{Aut}_Z(P_N)$, where the subscript means that the center is fixed pointwise.

The center of the Heisenberg image and the scalar subgroup of the Weil image need not coincide. Paull's affine and mod-4 models retain a μ_4-valued Heisenberg central character; see Section 5.2.

1.3 Dyadic conventions

Let $F/\mathbb{Q}_2$ be finite, with ring of integers $\mathcal{O}$ and uniformizer ϖ. Write

$$k = \mathcal{O}/\varpi\mathcal{O}, \quad |k| = q = 2^f, \quad e = v_F(2), \quad d = [F:\mathbb{Q}_2] = ef, \quad n \geq 1.$$

The valuation is normalized by $v_F(\varpi) = 1$. Set

$$R = \mathcal{O}/2\mathcal{O}, \qquad A = \mathcal{O}/4\mathcal{O}, \qquad D = n(2n+1).$$

Thus $|R| = q^e$ and $|A| = q^{2e}$. In the unramified case, $R = k$ and $A = W_2(k)$. In the ramified case, R has nilpotents and $A \neq \mathcal{O}/\varpi^2\mathcal{O}$.

Fix a smooth additive character $\psi : F \to \mathbb{C}^\times$ satisfying

$$\{a \in F : \psi(a\mathcal{O}) = 1\} = 4\mathcal{O}. \tag{1.2}$$

This is a condition on the conductor ideal. It cannot generally be replaced by $\ker\psi = 4\mathcal{O}$: if $d > 1$, then $(\mathcal{O}/4\mathcal{O},+) \simeq (\mathbb{Z}/4)^d$ cannot embed faithfully into a finite cyclic subgroup of $\mathbb{C}^\times$. Condition (1.2) agrees with [1, Notation, p. 2].

Take

$$W = F^{2n}, \quad L = \mathcal{O}^{2n} = L_X \oplus L_Y, \qquad \langle (x,y),(x',y')\rangle = x\cdot y' - x'\cdot y.$$

Let

$$p : \mathrm{Mp}(W) \to \mathrm{Sp}(W), \qquad K = \mathrm{Sp}(L), \qquad \widetilde{K} = p^{-1}(K).$$

Here $\mathrm{Mp}(W)$ denotes the twofold metaplectic cover. The symplectic rank of a free symplectic module of rank $2a$ means a; this differs from the rank of a Clifford representation defined in Section 8. Write ϵ for the nontrivial element of the covering kernel. A representation of this double cover is *genuine* if ϵ acts as $-I$; in particular, $\omega_\psi(\epsilon) = -I$. We write $\mu_r \subset \mathbb{C}^\times$ for the group of r-th roots of unity. For $r = 2, 4$, define

$$K(r) = \ker(K \to \mathrm{Sp}_{2n}(\mathcal{O}/r\mathcal{O})), \qquad \widetilde{K}(r) = p^{-1}(K(r)).$$

The numbers $2, 4$ always refer to ideals generated by rational integers. We separately write $K(\varpi) = \ker(K \to \mathrm{Sp}_{2n}(k))$.

Put

$$\mathcal{S}_L = \mathcal{S}(\mathcal{O}^n/2\mathcal{O}^n) \subset \mathcal{S}(F^n), \qquad \tau_L = \omega_\psi|_{\widetilde{K},\mathcal{S}_L}.$$

Here $\mathcal{S}(F^n)$ is the Schwartz–Bruhat space of locally constant, compactly supported functions. Thus $\mathcal{S}_L$ consists of functions supported on $\mathcal{O}^n$ and constant modulo $2\mathcal{O}^n$, with delta basis $\delta_x = \mathbf{1}_{x+2\mathcal{O}^n}$. We call τ_L the lattice representation (the *minimal type* of [1]). Throughout, a K-type means an irreducible representation occurring on restriction to a compact group K, without a Bernstein-type assertion.

2 The lattice model and descent modulo four

We use the local Heisenberg group

$$H(W) = W \times F, \qquad (v,t)(v',t') = (v+v', t+t'+\langle v, v'\rangle).$$

Its Schrödinger representation is

$$\rho_\psi(x,y,t)\phi(z) = \psi(t + 2x\cdot z - x\cdot y)\phi(z-y). \tag{2.1}$$

This convention for the central cocycle contains no factor of $1/2$; the corresponding Fourier kernel is $\psi(2x\cdot y)$. For the convention with central term $\frac{1}{2}\langle v, v'\rangle$, the corresponding character is $\psi'(a) = \psi(2a)$, not the same ψ. This gives the same long-root multiplier and Fourier kernel.

2.1 Canonical root-subgroup formulas

With the canonical lifts of [2, §§1.3–1.4], one has

$$\begin{aligned} \omega_\psi(\widetilde{x}_{\epsilon_i-\epsilon_j}(t))\phi(y) &= \phi(y + ty_i e_j), \\ \omega_\psi(\widetilde{x}_{\epsilon_i+\epsilon_j}(t))\phi(y) &= \psi(2ty_iy_j)\phi(y), \\ \omega_\psi(\widetilde{x}_{2\epsilon_i}(t))\phi(y) &= \psi(ty_i^2)\phi(y). \end{aligned} \tag{2.2}$$

The integral long-root phase is well defined on $\mathcal{S}_L$, since

$$t((y+2z)^2 - y^2) = 4t(yz + z^2) \in 4\mathcal{O}.$$

On this space, integral root operators depend only on $t \bmod 4\mathcal{O}$; the first two, corresponding to short roots, in fact depend only on $t \bmod 2\mathcal{O}$. Negative roots have the same property by conjugation with integral Weyl elements preserving the finite space.

The finite Fourier pairing is

$$\beta(x,y) = \psi(2x\cdot y), \qquad x, y \in R^n. \tag{2.3}$$

If $\beta(x,y) = 1$ for every y, condition (1.2) gives $2x \in 4\mathcal{O}^n$, hence $x = 0$ in R^n. Thus the pairing is perfect, and every entry of the normalized Fourier matrix is nonzero. Its Weil scalar prefactor may be an eighth root of unity and cannot simply be discarded in advance.

Proposition 2.1 (The lattice representation)**.** $\mathcal{S}_L$ is an irreducible $\widetilde{K}$-invariant subspace of dimension q^{en}.

Proof. Stability and irreducibility are established in [1, Theorem A.1]. Here is a direct check of irreducibility. Suppose $y, z \in \mathcal{O}$ satisfy $\psi(ty^2) = \psi(tz^2)$ for every $t \in \mathcal{O}$. Then $y^2 - z^2 \in 4\mathcal{O}$. If $v_F(y-z) < e$, the identity $y + z = (y-z) + 2z$ gives

$$v_F(y+z) = v_F(y-z) < e,$$

contradicting $v_F(y^2 - z^2) \geq 2e$. Consequently $y = z \bmod 2\mathcal{O}$. The long-root phases in each coordinate separate all delta basis vectors. Every commuting operator is therefore diagonal. Commuting also with the Fourier operator, all of whose entries are nonzero, forces it to be scalar. Since the lattice representation is unitary, it is irreducible. Its dimension is $|\mathcal{O}/2\mathcal{O}|^n = q^{en}$. □

2.2 A normal splitting over the level-four congruence subgroup

For any root, define

$$\widetilde{w}_\alpha(a) = \widetilde{x}_\alpha(a)\widetilde{x}_{-\alpha}(-a^{-1})\widetilde{x}_\alpha(a), \qquad \widetilde{h}_\alpha(a) = \widetilde{w}_\alpha(a)\widetilde{w}_\alpha(-1).$$

Inversion and the additive root-subgroup relations imply $\widetilde{w}_\alpha(-1) = \widetilde{w}_\alpha(1)^{-1}$. The mod-4 dependence of the root-subgroup operators therefore gives

$$a \in 1 + 4\mathcal{O} \quad \Longrightarrow \quad \tau_L(\widetilde{h}_\alpha(a)) = I. \tag{2.4}$$

Proposition 2.2 (The unique lift acting trivially)**.** Every $g \in K(4)$ has exactly one metaplectic lift acting as I on $\mathcal{S}_L$.

Proof. Write $g = \begin{pmatrix} A_0 & B_0 \\ C_0 & D_0 \end{pmatrix}$. Since $A_0 \equiv I \pmod{4\mathcal{O}}$, there is a block decomposition

$$g = \begin{pmatrix} I & 0 \\ C_0A_0^{-1} & I \end{pmatrix} \begin{pmatrix} A_0 & 0 \\ 0 & A_0^{-T} \end{pmatrix} \begin{pmatrix} I & A_0^{-1}B_0 \\ 0 & I \end{pmatrix}. \tag{2.5}$$

The parameter matrices in the two outer factors are symmetric and lie in $4M_n(\mathcal{O})$, so their root-subgroup lifts act as I on the finite space. Row elimination on A_0 has all off-diagonal elimination parameters in $4\mathcal{O}$ and all pivots in $1 + 4\mathcal{O}$. The corresponding short-root operators act trivially; the remaining torus operators act trivially by (2.4). Their product supplies the required lift. The other lift differs by ϵ and acts as $-I$, which proves uniqueness. □

It follows that

$$K^\sharp(4) = \ker \tau_L \cap \widetilde{K}(4)$$

projects isomorphically onto $K(4)$. As the intersection of two normal subgroups, it is normal in $\widetilde{K}$. Uniqueness makes the inverse of the projection a homomorphism, and smoothness and compactness make it a topological isomorphism. Integral symplectic reduction is surjective, so we obtain the exact sequence

$$1 \longrightarrow \mu_2 \longrightarrow E_{F,n} := \widetilde{K}/K^\sharp(4) \longrightarrow \mathrm{Sp}_{2n}(A) \longrightarrow 1, \tag{2.6}$$

and τ_L descends to a genuine irreducible representation $\overline{\tau}_L$.

Proposition 2.3 (Fixed vectors in the full local space)**.**

$$\mathcal{S}(F^n)^{K^\sharp(4)} = \mathcal{S}_L.$$

Proof. The inclusion of $\mathcal{S}_L$ in the left-hand side holds by definition. Conversely, all $\widetilde{x}_{\pm 2\epsilon_j}(4u)$, for $u \in \mathcal{O}$, belong to $K^\sharp(4)$. Invariance under the upper long roots implies that, whenever $\phi(y) \neq 0$, one has $\psi(4uy_j^2) = 1$ for every $u \in \mathcal{O}$. Thus $y_j^2 \in \mathcal{O}$, hence $y_j \in \mathcal{O}$, and the support lies in $\mathcal{O}^n$. Applying partial Fourier transforms to lower-long-root invariance shows that the support in every dual coordinate also lies in $\mathcal{O}$. For the kernel $\psi(2xy)$, the annihilator of $\mathcal{O}$ is $2\mathcal{O}$. The function is therefore $2\mathcal{O}$-periodic in each coordinate, so it belongs to $\mathcal{S}_L$. □

The finite descent is

$$\omega_\psi \xrightarrow{\text{restriction}} \omega_\psi|_{\widetilde{K}} \xrightarrow{K^\sharp(4)\text{-fixed vectors}} \mathcal{S}_L \xrightarrow{\text{finite quotient}} \overline{\tau}_L.$$

Here $E_{F,n}$ is the covering group and $G_{F,n} := \tau_L(\widetilde{K})$ its operator image. Taking $\widetilde{K}(4)$-invariants instead gives zero, because ϵ acts as $-I$.

Proposition 2.4 (Nonsplitting of the finite central extension)**.** The extension (2.6) is nonsplit, and its central kernel lies in the derived subgroup.

Proof. Work in a single coordinate. Choose $c \in \mathcal{O}$ with $v_F(c) = e-1$. The character $y \mapsto \psi(2cy)$ depends only on y mod $\varpi\mathcal{O}$. The map

$$\varpi^{2e-1}\mathcal{O}/4\mathcal{O} \longrightarrow \widehat{k}, \qquad b \longmapsto [y \mapsto \psi(by^2)]$$

is bijective: ψ is nontrivial on the last layer, the multiplicative pairing on k is nondegenerate, and squaring on k is bijective. Choose b such that $\psi(by^2) = \psi(2cy)$. The corresponding lower-long-root operator is the translation X_c. The sign in Fourier conjugation does not affect this conclusion, since the character takes values in $\{\pm 1\}$.

We can also choose $a \in \varpi\mathcal{O}$ with $\psi(ac^2) = -1$. Since $2acy \in 4\mathcal{O}$, the upper-long-root phase $T_a\phi(y) = \psi(ay^2)\phi(y)$ satisfies

$$T_a X_c = -X_c T_a.$$

On the other hand, $ab \in 4\mathcal{O}$, so the upper and lower unipotent matrices

$$u_+(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \qquad u_-(b) = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$$

commute in $\mathrm{SL}_2(A)$. The commutator of their lifts in $E_{F,n}$ therefore lies in the central kernel. It acts as $-I$, so it is the nontrivial kernel element. Changing lifts only multiplies them by central signs and does not change this commutator. The extension cannot split. □

The same commutator shows that the associated class in $H^2(\mathrm{Sp}_{2n}(A), \mathbb{C}^\times)$ is nonzero. A section of (2.6) represents it by a μ_2-valued cocycle.

3 Images of the Heisenberg and Weil representations

3.1 Heisenberg quotient and Pauli labels

Set

$$H_L = L \times \mathcal{O}, \qquad N_L = 2L \times 4\mathcal{O}.$$

The subgroup N_L is normal in H_L and acts trivially on $\mathcal{S}_L$. The center of H_L/N_L is A, with central character

$$\chi : A \to \mu_4, \qquad t \mapsto \psi(t).$$

This character is surjective because $\psi(4\mathcal{O}) = 1$ and $\psi(2\mathcal{O}) \neq 1$. It is not faithful when $d > 1$. We therefore quotient further by its central kernel:

$$P_{\mathrm{full}} = (H_L/N_L)/\ker\chi, \qquad Z(P_{\mathrm{full}}) = \mu_4, \quad |P_{\mathrm{full}}| = 4q^{2en}. \tag{3.1}$$

After choosing dual $\mathbb{F}_2$-bases, this is the standard dn-qubit Pauli group.

Write

$$\psi(2a) = (-1)^{\lambda(\bar{a})}, \qquad \lambda : R \to \mathbb{F}_2.$$

The pairing $\lambda(ab)$ on R is nondegenerate and $\mathbb{F}_2$-bilinear. Define

$$(X_c\phi)(y) = \phi(y-c), \qquad (Z_c\phi)(y) = (-1)^{\lambda(c\cdot y)}\phi(y), \quad c, y \in R^n. \tag{3.2}$$

Then

$$Z_a X_b = (-1)^{\lambda(a\cdot b)} X_b Z_a.$$

Local Weil covariance implies that $G_{F,n}$ normalizes P_{full} and induces on the label space R^{2n} the action of the underlying group modulo $2\mathcal{O}$. However, $G_{F,n}$ need not contain all of P_{full}.

Lemma 3.1 (The image of the adjoint of squaring)**.** Let $T : R \to R$ be the $\mathbb{F}_2$-linear adjoint of the squaring map, defined by

$$\lambda(T(a)x) = \lambda(ax^2) \qquad (a, x \in R).$$

Then

$$\operatorname{im} T = J := \varpi^{\lfloor e/2 \rfloor} R.$$

Proof. The kernel of the squaring map is $\varpi^{\lceil e/2 \rceil} R$. The multiplicative pairing satisfies $(\varpi^s R)^\perp = \varpi^{e-s} R$, the ideal-annihilator identity following from (1.2). Hence

$$\operatorname{im} T = (\ker(x \mapsto x^2))^\perp = \varpi^{\lfloor e/2 \rfloor} R.$$

□

An upper long-root element with parameter $2a$ acts as $Z_{T(a)}$, and the corresponding lower long-root element acts as $X_{T(a)}$. This explains the smaller set of Pauli labels in the congruence layer in the ramified case.

3.2 The congruence subgroup image, center, and kernel

Theorem 3.2 (Image of the level-two congruence subgroup)**.** Let

$$P_J = \langle \mu_{m_e} I, X(J^n), Z(J^n) \rangle, \qquad m_e = \begin{cases} 4, & e \text{ odd}, \\ 2, & e \text{ even}. \end{cases}$$

Then $\tau_L(\widetilde{K}(2)) = P_J$, and

$$|P_J| = m_e q^{2n\lceil e/2 \rceil}.$$

Proof. Apply (2.5) to an element of $K(2)$. The two symmetric parameters belong to $2M_n(\mathcal{O})$; their off-diagonal terms contribute phases with arguments in $4\mathcal{O}$, while their diagonal terms give $Z(J^n)$ and $X(J^n)$ by Lemma 3.1. The elimination parameters for the middle linear block belong to $2\mathcal{O}$ and act trivially on R^n. The remaining torus units have the form $a = 1 + 2t$, with operator $\kappa(a)$ acting as a scalar on the finite space.

By the multiplication law for the canonical torus lifts in [2, §3.1, p. 1117],

$$\widetilde{h}(a)^2 = (a, a)_F \widetilde{h}(a^2).$$

Since $a^2 \in 1 + 4\mathcal{O}$, equation (2.4) gives

$$\kappa(a)^2 = (a, a)_F = (a, -1)_F. \tag{3.3}$$

The norm projection formula for the Hilbert symbol and the formula for units in $\mathbb{Q}_2$ give

$$\begin{aligned} (a, -1)_F &= (N_{F/\mathbb{Q}_2} a, -1)_{\mathbb{Q}_2} \\ &= (-1)^{\operatorname{Tr}_{F/\mathbb{Q}_2}(t)}. \end{aligned} \tag{3.4}$$

The last step follows from

$$N(1 + 2t) = \det(I + 2M_t) \equiv 1 + 2\operatorname{Tr}(t) \pmod 4, \quad (u, -1)_{\mathbb{Q}_2} = (-1)^{(u-1)/2} \quad (u \in \mathbb{Z}_2^\times).$$

For the relevant norm compatibility and unit formula, see [8, Propositions 9.2.7, 9.3.6].

The ϖ-adic filtration of $\mathcal{O}/2\mathcal{O}$ has e layers isomorphic to k, and multiplication by t acts on each layer as multiplication by $\bar{t}$. Thus

$$\operatorname{Tr}_{F/\mathbb{Q}_2}(t) \bmod 2 = e \operatorname{Tr}_{k/\mathbb{F}_2}(\bar{t}). \tag{3.5}$$

If e is even, every $\kappa(a)$ is ± 1. If e is odd, surjectivity of the finite-field trace gives some $\kappa(a) = \pm i$. The covering kernel always supplies $-I$, and the Pauli commutation relations produce only signs. The upper and lower long-root elements supply all J-labels, proving the asserted equality. Operators with distinct Pauli labels do not differ by a scalar, so the order is $m_e|J|^{2n} = m_e q^{2n\lceil e/2\rceil}$. □

Theorem 3.3 (The exact sequence and center of the finite image)**.**

$$1 \longrightarrow P_J \longrightarrow G_{F,n} \longrightarrow \mathrm{Sp}_{2n}(R) \longrightarrow 1, \qquad Z(G_{F,n}) = \mu_{m_e} I.$$

Proof. If $\tau_L(\widetilde{g}) = I$, Heisenberg covariance implies that $g \bmod 2\mathcal{O}$ acts trivially on R^{2n}, so $g \in K(2)$. The reduction map therefore descends to $G_{F,n}$, with kernel given by Theorem 3.2. Surjectivity of integral reduction gives surjectivity at the end of the exact sequence. Likewise, every scalar comes from $\widetilde{K}(2)$, so the scalar subgroup is exactly $\mu_{m_e} I$. Since τ_L is irreducible, every central element is a scalar. □

For even e, $J^2 = 0$, so P_J is abelian and $Z(G_{F,n}) = \mu_2 I$, although P_{full} has center $\mu_4 I$.

Corollary 3.4 (Group orders and the representation kernel)**.**

$$\begin{aligned} |\mathrm{Sp}_{2n}(k)| &= q^{n^2} \prod_{j=1}^{n} (q^{2j} - 1), \\ |\mathrm{Sp}_{2n}(R)| &= q^{(e-1)D} |\mathrm{Sp}_{2n}(k)|, \\ |E_{F,n}| &= 2q^{(2e-1)D} |\mathrm{Sp}_{2n}(k)|, \\ |G_{F,n}| &= m_e q^{2n\lceil e/2\rceil + (e-1)D} |\mathrm{Sp}_{2n}(k)|. \end{aligned} \tag{3.6}$$

Moreover,

$$\ker \overline{\tau}_L \simeq (C_2)^r, \qquad r = f\big(eD - 2n\lceil e/2\rceil\big) - \mathbf{1}_{\{e \text{ is odd}\}}. \tag{3.7}$$

Proof. Smoothness of the integral symplectic group implies that each additional layer of ϖ-adic precision has a reduction kernel of order q^D, giving the order formulas. Genuineness implies that the representation kernel intersects the covering kernel trivially. The representation kernel therefore projects injectively into the additive group of

$$K(2)/K(4) \simeq \mathfrak{sp}_{2n}(R)$$

which has exponent 2. Taking the quotient of the group orders gives (3.7). □

3.3 Restriction to the level-two congruence subgroup

Theorem 3.5 (Multiplicity-free decomposition)**.** The restriction $\tau_L|_{\widetilde{K}(2)}$ decomposes without multiplicities into $q^{2n\lfloor e/2\rfloor}$ irreducible representations, each of dimension $q^{n(e \bmod 2)}$.

Proof. The radical of the multiplicative pairing on J is

$$B = J^{\perp} = \varpi^{\lceil e/2\rceil} R \subseteq J.$$

Since $B^2 = 0$, the subgroups $X(B^n)$ and $Z(B^n)$ generate an elementary abelian group of order $|B|^{2n} = q^{2n\lfloor e/2\rfloor}$ that commutes with P_J. Every nonidentity element of this abelian group has a nonzero Pauli label and hence trace zero. Each joint character therefore occurs with multiplicity

$$\frac{q^{en}}{|B|^{2n}} = q^{n(e \bmod 2)}.$$

On the corresponding joint eigenspace, the alternating pairing on the quotient labels $(J/B)^{2n}$ is nondegenerate. The dimension of an irreducible finite Heisenberg representation is precisely the dimension above, so each joint eigenspace is irreducible. Distinct central characters distinguish these constituents, making the decomposition multiplicity-free. □

3.3.1 Multiplicity spaces in the unramified case

Put $N = \widetilde{K}(2)$. If $e = 1$, let $\eta = \tau_L|_N$. Theorem 3.5 ensures that η is irreducible. For a smooth $\widetilde{K}$-representation (π, V), set

$$M_{\tau_L}(V) = \mathrm{Hom}_N(\eta, V), \qquad (\widetilde{g} \cdot T) = \pi(\widetilde{g}) T \tau_L(\widetilde{g})^{-1}.$$

This action is trivial on N, so $M_{\tau_L}(V)$ is a representation of $\widetilde{K}/N \simeq \mathrm{Sp}_{2n}(k)$. The evaluation map gives

$$V[\eta] \simeq \tau_L \otimes \mathrm{Inf}\, M_{\tau_L}(V), \tag{3.8}$$

where $V[\eta]$ is the η-isotypic part of the restriction to N. This is the isotypic decomposition for the compact group N; τ_L corresponds to the trivial one-dimensional multiplicity space.

3.3.2 The obstruction to direct reduction and the ramified case

The full finite Weil space considered here does not descend projectively along $K \to \mathrm{Sp}_{2n}(k)$: the subgroup $K(2) \subseteq K(\varpi)$ already contains elements acting as nonscalar X_c, Z_c, whereas a projective representation factoring through this reduction must have the entire reduction kernel act by scalars. This also holds in the unramified case, where the full Pauli action is the obstruction.

When $e > 1$, the restriction $\tau_L|_N$ is itself reducible and thus cannot still be treated as a single irreducible representation in (3.8). One must choose an irreducible constituent η_0, determine its inertia subgroup

$$I_{\eta_0} = \{\widetilde{g} \in \widetilde{K} : \eta_0^{\widetilde{g}} \simeq \eta_0\},$$

and then analyze extensions and the intertwining cocycle. These inertia groups and extension cocycles remain to be determined; see Section 18.

4 Dyadic examples

4.1 The $\mathbb{Q}_2$ Clifford group

When $F = \mathbb{Q}_2$, we have $e = f = 1$, $R = \mathbb{F}_2$, and $A = \mathbb{Z}/4$. Write $E_n = E_{\mathbb{Q}_2,n}$ and $G_n = G_{\mathbb{Q}_2,n}$. Here $J = R$, so

$$1 \longrightarrow P_n \longrightarrow G_n \longrightarrow \mathrm{Sp}_{2n}(\mathbb{F}_2) \longrightarrow 1, \qquad Z(G_n) = \mu_4 I. \tag{4.1}$$

The conjugation action also gives

$$G_n/\mu_4 \simeq \mathrm{Aut}_Z(P_n), \qquad \mathrm{Cl}_n = U(1)G_n, \qquad \mathrm{Cl}_n\,/U(1) \simeq G_n/\mu_4.$$

Here Aut_Z denotes the group of automorphisms fixing the center pointwise. Indeed, the inner automorphisms of P_n give all central phase corrections that induce the identity on P_n/μ_4, G_n realizes all symplectic transformations on the labels, and Schur's lemma implies that unitary operators implementing the same automorphism differ only by a scalar. No assertion is made here that (4.1) or its projective quotient is necessarily a semidirect product.

4.1.1 Normalization of the Weyl operator

Take $\psi(1) = i$, and let

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \zeta_8 = e^{\pi i/4}.$$

The canonical root-subgroup lift gives S, and the canonical Weyl element gives

$$W = SHSHS = \zeta_8 H = \frac{1+i}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \qquad W^2 = iI. \tag{4.2}$$

For an odd unit a, the root-subgroup word and the congruence $a^{-1} \equiv a \pmod 4$ give

$$\tau_L(\widetilde{w}(a)) = \begin{cases}\zeta_8 H & a \equiv 1 \pmod 4,\\ \zeta_8^{-1} H & a \equiv -1 \pmod 4,\end{cases} \qquad \tau_L(\widetilde{h}(a)) = \begin{cases}I & a \equiv 1 \pmod 4,\\ -iI & a \equiv -1 \pmod 4.\end{cases}$$

Replacing W by H changes the linear group and its phases. The scalar center is μ_4 by Theorem 3.3; the generators $S_j, W_j, \mathrm{CNOT}, CZ$ have entries in $\mathbb{Q}(i)$.

4.1.2 The covering kernel and the representation kernel

The general order formulas now become

$$|E_n| = 2^{n(2n+1)+1}|\operatorname{Sp}_{2n}(\mathbb{F}_2)|, \qquad |G_n| = 2^{2n+2}|\operatorname{Sp}_{2n}(\mathbb{F}_2)|, \qquad \ker\overline{\tau}_L \simeq (C_2)^{2n^2-n-1}.$$

Proposition 4.1. The projection of $\ker\overline{\tau}_L$ to $\operatorname{Sp}_{2n}(\mathbb{Z}/4)$ is

$$\left\{ I + 2T :\ T = \begin{pmatrix}a & b\\ c & a^T\end{pmatrix},\quad b = b^T,\ c = c^T,\ \operatorname{diag} b = \operatorname{diag} c = 0,\ \operatorname{tr} a = 0 \right\},$$

where all blocks have entries in $\mathbb{F}_2$. The projection is injective on the representation kernel.

Proof. A kernel element must be trivial modulo 2. The symplectic condition gives the displayed block form. In the Gauss decomposition of $I + 2T$, the finite operators of the upper and lower unipotent blocks are, up to scalars, $Z^{\operatorname{diag} b}$ and $X^{\operatorname{diag} c}$, respectively. Thus the operator is scalar if and only if both diagonal vectors vanish. In this case, diagonal elimination in the linear block gives $(-i)^r$, where r is the number of diagonal units congruent to 3 (mod 4). A lift acting as I exists if and only if r is even, equivalently,

$$\det(I + 2a) \equiv 1 \pmod 4 \quad \Longleftrightarrow \quad \operatorname{tr} a = 0.$$

Genuineness ensures that each such element of the underlying group has exactly one lift in the representation kernel. The displayed linear space has dimension $n^2 - 1 + 2 \cdot n(n-1)/2 = 2n^2 - n - 1$. □

n	$\dim \mathcal{S}_L$	$\lvert E_n\rvert$	$\lvert G_n\rvert$	$\lvert\ker\overline{\tau}_L\rvert$
1	2	96	96	1
2	4	1 474 560	46 080	32

Thus, starting with two qubits, the double cover modulo 4 already differs from its actual Clifford matrix image.

4.2 Unramified extensions and Witt rings of length two

When $e = 1$, we have $R = k$, $A = W_2(k)$, and

$$1 \longrightarrow P_{nf} \longrightarrow G_{F,n} \longrightarrow \operatorname{Sp}_{2n}(k) \longrightarrow 1.$$

After choosing dual $\mathbb{F}_2$-bases, this is a finite subgroup of the nf-qubit Clifford group. If $f > 1$, it realizes only the symplectic actions preserving the k-linear structure:

$$\operatorname{Sp}_{2n}(k) \subsetneq \operatorname{Sp}_{2nf}(\mathbb{F}_2).$$

Its projective image is therefore a proper subgroup of $\mathrm{Cl}_{nf}/U(1)$ when $f > 1$.

Gurevich–Hadani [3, §1.1] obtain $W_2(k)$ from an unramified dyadic extension and retain a central character with values in the fourth roots of unity. This corresponds to the finite Heisenberg data in our construction. After matching the group action, polarization, and central character, Stone–von Neumann uniqueness identifies the projective intertwining operators. This uniqueness determines operators only up to a scalar; comparing specified linear covers still requires checking the cocycle values and normalizations.

4.3 The ramified quadratic field $\mathbb{Q}_2(\sqrt{2})$

Take $F = \mathbb{Q}_2(\varpi)$ with $\varpi^2 = 2$, so $e = 2, f = 1$, and

$$R \simeq \mathbb{F}_2[\varepsilon]/(\varepsilon^2), \qquad J = \varepsilon R.$$

Let ψ_0 have kernel $4\mathbb{Z}_2$ on $\mathbb{Q}_2$, with $\psi_0(1) = i$. The different ideal is generated by 2ϖ, so we may take

$$\psi(a + b\varpi) = \psi_0(b) = \psi_0\left(\mathrm{Tr}_{F/\mathbb{Q}_2} \frac{a + b\varpi}{2\varpi}\right).$$

This satisfies the conductor-ideal condition used in this note. Writing the finite coordinate as $x = a + b\varepsilon$, with $a, b \in \mathbb{F}_2$, gives $\lambda(x) = b$.

For $n = 1$, order the basis by $(a, b) = (0,0), (0,1), (1,0), (1,1)$. Thus the basis is $(\delta_0, \delta_\varpi, \delta_1, \delta_{1+\varpi})$, where $\delta_r(s) = 1$ if $s = r$ in R, and is zero otherwise. In the local function space these vectors correspond, up to a common normalization, to the characteristic functions of $r + 2\mathcal{O}$. The upper long-root operators with parameters $1, \varpi$ and the finite Fourier matrix are

$$D_1 = \mathrm{diag}(1, 1, 1, -1), \qquad D_\varpi = \mathrm{diag}(1, -1, i, -i), \qquad H_{(a,b),(c,d)} = \frac{1}{2}(-1)^{ad+bc}.$$

Here $H^2 = I$, and

$$D_1 H D_1 H D_1 = H.$$

The canonical Weyl operator in this example is therefore exactly H. This phase is computed from the root-subgroup word and cannot be imported directly from the $\mathbb{Q}_2$ case. The group $\langle D_1, D_\varpi, H\rangle$ gives the actual finite image, and

$$|E_{F,1}| = 6144, \quad |G_{F,1}| = 384, \quad Z(G_{F,1}) = \mu_2 I, \quad |\ker \overline{\tau}_L| = 16.$$

The image of its congruence kernel modulo 2 is

$$P_J = \langle -I, X_\varepsilon, Z_\varepsilon\rangle \simeq (C_2)^3.$$

Thus P_J is abelian, while the full Pauli group has center μ_4 and the Weil image has center μ_2. Section 6.6 substitutes these three generators into Paull's intertwiner formula and compares both their phases and the resulting finite covering group.

F, with $n = 1$ throughout	$\dim \mathcal{S}_L$	$\lvert E_{F,1}\rvert$	$\lvert G_{F,1}\rvert$	$\lvert Z(G_{F,1})\rvert$
$\mathbb{Q}_2$	2	96	96	4
Unramified quadratic extension	4	7680	3840	4
$\mathbb{Q}_2(\sqrt{2})$	4	6144	384	2

The equal representation dimensions in the last two rows do not determine the cover, image or center. Exact matrix checks are recorded in Appendix C.

5 Restriction of scalars and Paull's models

Put $m = nd$, the symplectic rank after restriction of scalars to $\mathbb{Q}_2$. The underlying free $\mathbb{Z}/4$-module in Paull's model has rank $2m$.

5.1 The inverse different and an integral symplectic basis

Suppose that $\mathfrak{D}_{F/\mathbb{Q}_2} = \delta\mathcal{O}$. Fix ψ_0 as above. Every ψ satisfying the conductor condition of this note can be written as

$$\psi(x) = \psi_0(\ell(x)), \qquad \ell(x) = \mathrm{Tr}_{F/\mathbb{Q}_2}(u\delta^{-1}x), \qquad u \in \mathcal{O}^\times. \tag{5.1}$$

By the definition of the inverse different, $(a, b) \mapsto \ell(ab)$ is a perfect $\mathbb{Z}_2$-pairing on $\mathcal{O}$. Consequently,

$$b_0(v, w) = \ell(\langle v, w\rangle_F)$$

makes L a self-dual symplectic $\mathbb{Z}_2$-lattice of rank $2m$. Choosing dual integral bases compatible with $L = L_X \oplus L_Y$ gives

$$\begin{aligned} &(L/4L, \bar{b}_0) \simeq ((\mathbb{Z}/4)^{2m}, b_{\mathrm{std}}), \\ &\iota : \mathrm{Sp}_{2n}(F) \hookrightarrow \mathrm{Sp}_{2m}(\mathbb{Q}_2), \qquad \iota_4 : \mathrm{Sp}_{2n}(A) \hookrightarrow \mathrm{Sp}_{2m}(\mathbb{Z}/4). \end{aligned} \tag{5.2}$$

The map ι_4 is injective: an identity action on the underlying $\mathbb{Z}/4$-module already forces the original A-linear matrix to be the identity. These identifications depend on ψ and the dual bases; changing the bases changes only the coordinate expression of the comparison.

Proposition 5.1 (Local descent is compatible with restriction of scalars)**.** As covers and specified representations, we have

$$E_{F,n} \simeq \iota_4^* E_{\mathbb{Q}_2,m}, \qquad \overline{\tau}_L \simeq \iota_4^* \overline{\tau}_{\mathbb{Q}_2,m}.$$

Proof. First identify the Heisenberg representations at the local level using the map $t \mapsto \ell(t)$ on the centers. The corresponding Schrödinger spaces simply use different coordinates for the same underlying locally compact additive group. The $\mathbb{C}^\times$-extension of the F-symplectic group by intertwining operators is therefore the pullback of the analogous extension of the larger $\mathbb{Q}_2$-symplectic group. By the construction of the local metaplectic group as a derived subgroup [2, §1.3], the derived subgroup of the smaller extension is contained in that of the larger extension. Both give double covers over the smaller symplectic group, so this inclusion is an equality on each fiber. This also identifies the Weil operators.

The integral lattice and the polarization remain unchanged, and

$$K_F(4) = K_F \cap K_{\mathbb{Q}_2,m}(4).$$

On both sides, the image of the splitting is characterized by "the unique lift acting as I on the corresponding finite space," so the splittings are compatible. Quotienting by these images and using the surjectivity of integral symplectic reduction gives the pullback isomorphism of finite covers and representations. □

5.2 Paull's finite Heisenberg models

The center-C_2 and affine center-C_4 models of Section 1 correspond to Paull's pseudosymplectic and affine constructions. The latter linearizes on a μ_4-extension $\mathrm{AMp}(B_W)$. The model directly relevant to descent modulo four instead starts from a free symplectic module $\widetilde{W} \simeq (\mathbb{Z}/4)^{2m}$ and

$$H_P = \widetilde{W} \times \mathbb{Z}/4, \qquad (v, t)(w, s) = (v + w, t + s + b_{\mathrm{std}}(v, w)). \tag{5.3}$$

The group $\mathrm{Sp}(\widetilde{W})$ acts directly on H_P, whose full center is $2\widetilde{W} \times \mathbb{Z}/4$. The chosen Schrödinger representation kills $2\widetilde{W} \times \{0\}$, has central character $s \mapsto i^s$, and has dimension 2^m.

5.3 Thesis versions and citation conventions

We distinguish the 371-page version [4] from the 376-page university-hosted version [5]; page references are printed page numbers. The latter proves rank-two existence by stabilization in Corollary 2.62(ii), pp. 193–194, whereas the former leaves it open in Remark 2.62, p. 188. The cocycle existence result is Corollary 2.61, p. 183, in [4], and Corollary 2.62, p. 188, in [5]. The comparison results retain their numbers: Lemmas 2.71–2.72 and Proposition 2.73 occur on pp. 198–203 and pp. 204–209, respectively.

Unless specified otherwise, equations (97), (109), (113), (203), (368), and (369), and Chapter 3 references use [4]. Rank-two existence and Section 6.6 use [5].

6 Comparison with Paull's model modulo 4

6.1 Identification of the Heisenberg representations

First consider

$$H_A = (L/4L) \times A, \qquad (v,t)(w,s) = (v+w, t+s+\langle v,w\rangle_F).$$

Let $\overline{\ell} : A \to \mathbb{Z}/4$ be the reduction of (5.1). Then

$$q_H : H_A \longrightarrow H_P, \qquad q_H(v,t) = (v, \overline{\ell}(t)) \tag{6.1}$$

is a surjective group homomorphism with kernel $\{0\} \times \ker \overline{\ell}$. Surjectivity follows from $\ell(\mathcal{O}) = \mathbb{Z}_2$. In the dual coordinates of §5, the local formula becomes

$$\rho_P(X,Y,s)f(Z) = i^{s+2X\cdot Z - X\cdot Y} f(Z-Y), \qquad Z \in \mathbb{F}_2^m. \tag{6.2}$$

Here $X, Y \in (\mathbb{Z}/4)^m$, and translation is interpreted modulo 2. This is precisely the corresponding Schrödinger model with central character $\chi_1(s) = i^s$ in Definition 2.46 and Definition–Proposition 2.51 of the uploaded version.

If $\Theta : \mathcal{S}_L \to \mathbb{C}[\mathbb{F}_2^m]$ is the unitary identification of the normalized delta bases, then

$$\Theta \rho_F(h) \Theta^{-1} = \rho_P(q_H(h)), \qquad q_H(gh) = \iota_4(g) q_H(h). \tag{6.3}$$

These identities identify the Heisenberg representations and their symplectic actions, including in the ramified case.

6.2 Projective comparison and scalar normalization

Theorem 6.1 (Projective comparison)**.** With the data above, as homomorphisms $\mathrm{Sp}_{2n}(A) \to PU(2^m)$, we have

$$[\Theta \overline{\tau}_L \Theta^{-1}] = \iota_4^*[\omega_m^P].$$

Proof. Choose a lift over g in the cover obtained by local descent, and denote its operator by $U_F(g)$. Write $U_P(\iota_4(g))$ for Paull's corresponding unitary intertwining operator. Equation (6.3) shows that the two operators implement the same automorphism of the same irreducible Heisenberg representation. Schur's lemma gives a unique $b(g) \in U(1)$ such that

$$\Theta U_F(g) \Theta^{-1} = b(g) U_P(\iota_4(g)). \tag{6.4}$$

Taking projective classes proves the assertion. □

If the cocycles of the two sections are c_F, c_P, respectively, then

$$c_F(g,h) = b(g)b(h)b(gh)^{-1} c_P(\iota_4(g), \iota_4(h)). \tag{6.5}$$

Once the matrices are given, the phase can be computed directly:

$$b(g) = 2^{-m} \operatorname{Tr}\Big(\Theta U_F(g) \Theta^{-1} U_P(\iota_4(g))^* \Big).$$

6.3 Linear comparison in symplectic rank at least three

Let Mp_m^P denote the double cover given by one of Paull's μ_2-valued normalizations matching the central character above, and let $\widetilde{\omega}_m^P$ be the specified Weil representation. For the rank hypotheses and thesis versions, see Section 5.3.

Theorem 6.2 (Simultaneous comparison of covers and representations)**.** If $m = nd \geq 3$, then

$$(E_{F,n}, \overline{\tau}_L) \simeq \iota_4^*(\mathrm{Mp}_m^P, \widetilde{\omega}_m^P). \tag{6.6}$$

The isomorphism fixes the base group and the central μ_2, and preserves the specified representations under Θ. The conclusion holds for every valid μ_2-valued normalization.

Proof. First compare $E_{\mathbb{Q}_2,m}$ and Mp_m^P over $G_0 = \mathrm{Sp}_{2m}(\mathbb{Z}/4)$. Both embed in $G_0 \times U(2^m)$, with fibers over g given by $\{(g, \pm U_{\mathrm{loc}}(g))\}$ and $\{(g, \pm U_P(g))\}$, respectively. By Theorem 6.1, $U_{\mathrm{loc}}(g) = b(g)U_P(g)$. Both cocycles take values in μ_2, so $\partial(b^2) = 1$; that is, b^2 is a linear character of G_0. For $m \geq 3$, the group G_0 is perfect [4, Proposition 2.7.v], so $b^2 = 1$. Thus the fibers on the two sides agree pointwise, identifying the covers and the operators simultaneously. Pull back along ι_4 and apply Proposition 5.1. □

For related Schur-multiplier results, see [6].

6.4 Comparison by stabilization in every symplectic rank

For $M \geq \max(m, 3)$, define

$$j_M(g) = \iota_4(g) \oplus I_{2(M-m)}.$$

Theorem 6.3 (Comparison by stabilization)**.** For every $F/\mathbb{Q}_2$ and $n \geq 1$, we have

$$E_{F,n} \simeq j_M^* \, \mathrm{Mp}_M^P, \qquad j_M^* \widetilde{\omega}_M^P \simeq \overline{\tau}_L \otimes I_{2^{M-m}}. \tag{6.7}$$

Proof. Decompose the symplectic space over $\mathbb{Q}_2$ as $W_M = W_m \perp W_{M-m}$. The Schrödinger space decomposes accordingly as a tensor product. There is an injective homomorphism between the extensions by intertwining operators,

$$(g, T) \longmapsto (g \oplus 1, T \otimes I).$$

Passing to derived subgroups maps the smaller metaplectic group into the larger one; counting fibers of the double covers shows that it is precisely the pullback. Its operators are exactly $T \otimes I$, with no undetermined additional phase. The finite lattice spaces also satisfy $\mathcal{S}_M = \mathcal{S}_m \otimes \mathcal{S}_{M-m}$, and the level-4 congruence subgroups and the images of the unique splittings acting trivially are compatible. The result therefore descends to finite covers. Apply Theorem 6.2 in symplectic rank M, followed by Proposition 5.1. □

The multiplicity in (6.7) is 2^{M-m}, so stabilization does not preserve multiplicity one. It also underlies Paull's rank-two existence proof [5].

6.5 The phase obstruction in low rank and a rank-one counterexample

For any finite group G, suppose that two systems of unitary intertwining operators satisfy

$$U_1(g) = b(g)U_2(g), \qquad c_1, c_2 \in Z^2(G, \mu_2).$$

Then $b^2 \in \mathrm{Hom}(G, U(1))$. With the base group and the central μ_2 fixed, the two extensions are isomorphic if and only if b^2 has a square root in the character group. More precisely,

$$\partial b = \partial a, \ a : G \to \mu_2 \iff b/a = \chi \text{ is a character} \iff b^2 = \chi^2. \tag{6.8}$$

For the reverse implication in the last equivalence, take $a = b/\chi$. Even if the extensions are isomorphic, the representations may still differ by the character χ. If the operators are required to agree pointwise under the fixed Heisenberg identification, one must require $b(g) \in \mu_2$. Thus projective agreement, an isomorphism of covers, and compatibility of specified linear representations are three distinct levels of comparison.

Proposition 3.51 of the uploaded version gives a concrete distinction: choice (368) yields $U_2(\mathbb{F}_3)$ and a faithful two-dimensional Weil representation; choice (369) yields $Q_8 \rtimes \mathrm{Dic}_3$, with representation kernel of order 2 and actual image of order 48. For the cover in this note with $\mathbb{Q}_2, n = 1$, the lifts of the base-group element $-I$ act as $\pm iI$, so they have order 4; in the latter specified model, $(-I, I)$ has order 2. It therefore cannot be directly identified with the cover in this note. Here $U_2(\mathbb{F}_3)$ is the usual notation for the finite unitary group.

6.6 An explicit comparison for the ramified quadratic example

Take $F = \mathbb{Q}_2(\varpi)$, $\varpi^2 = 2$, $n = 1$, with the character and basis of Section 4.3. Since $m = 2$, we use Paull's rank-two formulas on $(\mathbb{Z}/4)^4$, with a four-dimensional Schrödinger representation.

Here $\ell(a + b\varpi) = b$. Choose the dual coordinates

$$x = \varpi X_1 + X_2, \qquad y = Y_1 + \varpi Y_2, \qquad \ell(xy) = X_1Y_1 + X_2Y_2.$$

The resulting coordinates on $\widetilde{W}$ are (X_1, X_2, Y_1, Y_2), with pairing

$$\langle (X, Y), (X', Y') \rangle = X \cdot Y' - X' \cdot Y.$$

In particular, the coordinate $z = (a, b) \in \mathbb{F}_2^2$ of the Schrödinger space corresponds to $a + b\varpi \in R$, with the same ordered delta basis as in the local calculation. Write

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For $t = s + r\varpi \in A$, direct multiplication gives

$$\iota_4(u(t)) = \begin{pmatrix} I_2 & B_t \\ 0 & I_2 \end{pmatrix}, \qquad B_t = \begin{pmatrix} r & s \\ s & 2r \end{pmatrix}. \tag{6.9}$$

Consequently, the three elements to compare are

$$g_1 = \begin{pmatrix} I_2 & J_0 \\ 0 & I_2 \end{pmatrix}, \quad g_\varpi = \begin{pmatrix} I_2 & \mathrm{diag}(1, 2) \\ 0 & I_2 \end{pmatrix}, \quad g_w = \begin{pmatrix} 0 & J_0 \\ -J_0 & 0 \end{pmatrix}. \tag{6.10}$$

They belong to $S = \iota_4(\mathrm{SL}_2(A)) \subset \mathrm{Sp}_4(\mathbb{Z}/4)$.

We use ρ_P from (6.2), with central character $\chi_1(t) = i^t$, and abbreviate $\rho_P(v) = \rho_P(v, 0)$. The source for the following formulas is the university-hosted version [5, Definition–Proposition 2.52, Lemma 2.54, and Definition–Proposition 2.56], specifically its formulas (201), (207), (215), and (218). For $g \in \mathrm{Sp}(\widetilde{W})$, put

$$K(g) = \ker(1 - g), \qquad I(g) = \mathrm{im}(1 - g), \qquad K_1(g) = \{v \in \widetilde{W} : 2v \in K(g)\}.$$

Choose $w_g \in \widetilde{W}$ satisfying

$$\langle gv, v \rangle = 2\langle w_g, v \rangle \pmod 4 \qquad (v \in K_1(g)). \tag{6.11}$$

Paull's full sum is

$$U_P(g) = \frac{\gamma_g}{|K(g)|\,|I(g) \cap 2\widetilde{W}|} \sum_{v \in \widetilde{W}} \rho_P(gv)^{-1} \rho_P(w_g) \rho_P(v), \qquad |\gamma_g|^2 = |2I(g)|^{-1}. \tag{6.12}$$

Equivalently, choose representatives S_g of $I(g)/(I(g)\cap 2\widetilde{W})$, and for each $u\in S_g$ choose v with $u=(1-g)v$. Then

$$U_P(g)=\gamma_g\sum_{u\in S_g} i^{\langle u,v+w_g\rangle+2\langle w_g,v\rangle}\rho_P(u+w_g). \tag{6.13}$$

All exponents are computed modulo four.

Let e_1,e_2 be the X-basis vectors and f_1,f_2 the Y-basis vectors. The choices that recover the local operators are

g	w_g	γ_g	$\lvert K(g)\rvert$	$\lvert I(g)\cap 2\widetilde{W}\rvert$	$\lvert 2I(g)\rvert$	$U_P(g)$
g_1	0	$1/2$	16	4	4	D_1
g_ϖ	e_2	$(1+i)/2$	32	4	2	D_ϖ
g_w	0	$1/2$	4	16	4	H

To verify (6.11), note that for g_1, membership in $K_1(g_1)$ forces Y to be even, and $\langle g_1v,v\rangle = 2Y_1Y_2=0 \pmod 4$. For g_ϖ, it forces Y_1 to be even, so

$$\langle g_\varpi v,v\rangle = Y_1^2+2Y_2^2 = 2Y_2 = 2\langle e_2,v\rangle \pmod 4.$$

In particular, $w_{g_\varpi}=0$ would fail for $v=f_2$. For g_w, the condition is $X\equiv J_0Y \pmod 2$, which gives $\langle g_wv,v\rangle = 2Y_1Y_2+2X_1X_2=0 \pmod 4$. Solving the linear congruences gives the table and $|\gamma_g|^2=|2I(g)|^{-1}$.

For g_1, take $u=ae_1+be_2$, $a,b\in\{0,1\}$, with $v=-bf_1-af_2$. The phase in (6.13) is $i^{-2ab}=(-1)^{ab}$. Set

$$Z_1=\rho_P(e_1)=\mathrm{diag}(1,1,-1,-1),\qquad Z_2=\rho_P(e_2)=\mathrm{diag}(1,-1,1,-1).$$

The short sum becomes

$$U_P(g_1)=\tfrac12(I+Z_1+Z_2-Z_1Z_2)=\mathrm{diag}(1,1,1,-1)=D_1. \tag{6.14}$$

For g_ϖ, one has

$$I(g_\varpi)=\{(a,2b,0,0): a\in\mathbb{Z}/4,\ b\in\mathbb{Z}/2\}.$$

Choose $S_{g_\varpi}=\{0,e_1\}$, with preimages $v=0,-f_1$, respectively. Since $w_{g_\varpi}=e_2$, the exponents in the short sum are

$$0,\qquad \langle e_1,-f_1+e_2\rangle+2\langle e_2,-f_1\rangle=-1.$$

It follows that

$$\begin{aligned}U_P(g_\varpi)&=\frac{1+i}{2}(\rho_P(e_2)-i\rho_P(e_1+e_2))\\&=\frac{1+i}{2}(I-iZ_1)Z_2=\mathrm{diag}(1,-1,i,-i)=D_\varpi.\end{aligned} \tag{6.15}$$

For g_w, representatives and preimages may be chosen as

$$u=(a,b,b,a),\qquad v=(a,b,0,0),\qquad a,b\in\{0,1\}.$$

Here $\langle u,v\rangle=-2ab$. Combining this phase with (6.2) yields

$$\begin{aligned}(U_P(g_w)f)(z)&=\frac12\sum_{a,b\in\mathbb{F}_2}(-1)^{az_1+bz_2}f(z_1-b,z_2-a)\\&=\frac12\sum_{z'\in\mathbb{F}_2^2}(-1)^{z_1z_2'+z_2z_1'}f(z').\end{aligned}$$

Thus

$$U_P(g_w) = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = H. \tag{6.16}$$

If $H_2 = 2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, then $H = \mathrm{SWAP}(H_2 \otimes H_2)$. The exchange of coordinates comes from J_0 in g_w. In these bases and normalizations, the phase $b(g)$ in (6.4) equals one on all three chosen generators.

To compare the covering groups, retain the base-group element as well as its matrix. Define

$$E^{\mathrm{pair}} = \langle (g_1, D_1), (g_\varpi, D_\varpi), (g_w, H) \rangle \subset S \times U(4). \tag{6.17}$$

There is a useful relation which detects the central sign. Already over F,

$$u(\varpi)w = \begin{pmatrix} -\varpi & 1 \\ -1 & 0 \end{pmatrix}, \qquad (u(\varpi)w)^2 = \begin{pmatrix} 1 & -\varpi \\ \varpi & -1 \end{pmatrix}, \qquad (u(\varpi)w)^4 = -I_2 = w^2.$$

The corresponding identities in the two four-dimensional spaces are

$$(g_\varpi g_w)^4 = g_w^2 = -I_{\widetilde{W}}, \qquad (D_\varpi H)^4 = -I_{\mathcal{S}_L}, \qquad H^2 = I_{\mathcal{S}_L}. \tag{6.18}$$

Hence E^{pair} contains $(1, -I_{\mathcal{S}_L})$. Nonsplitting follows from Proposition 2.4, not from this relation for a single choice of lifts.

Proposition 6.4 (The covering group in the worked comparison)**.** The map

$$E_{F,1} \longrightarrow S \times U(4), \qquad \widetilde{g} \longmapsto (\iota_4(\mathrm{pr}(\widetilde{g})), \overline{\tau}_L(\widetilde{g}))$$

identifies $E_{F,1}$ with E^{pair}. In particular,

$$1 \longrightarrow \mu_2 \longrightarrow E^{\mathrm{pair}} \longrightarrow S \longrightarrow 1,$$

and

$$|S| = 3072, \qquad |E^{\mathrm{pair}}| = 6144, \qquad |\operatorname{im}(E^{\mathrm{pair}} \to U(4))| = 384, \qquad |\ker(E^{\mathrm{pair}} \to U(4))| = 16.$$

Proof. The map is injective: an element trivial in both coordinates belongs to the central μ_2, on which the genuine representation is faithful. The operators in Section 4.3 are operators of specified local lifts, so the image contains the three displayed pairs. The elements $u(1), u(\varpi), w$ generate $\mathrm{SL}_2(A)$: the first two generate all upper unipotents, since A is additively generated by $1, \varpi$, and conjugation by w gives all lower unipotents. Elementary row operations generate SL_2 over the local ring A. Thus E^{pair} surjects onto S; by (6.18), it also contains the covering kernel. It is consequently the full image of $E_{F,1}$.

Since $4\mathcal{O} = \varpi^4\mathcal{O}$, successive reduction to $\mathbb{F}_2$ gives

$$|\mathrm{SL}_2(A)| = |\mathrm{SL}_2(\mathbb{F}_2)|\, 2^{3(4-1)} = 6 \cdot 2^9 = 3072.$$

The cover has twice this order. The image order 384 was computed in Section 4.3; the representation kernel therefore has order $6144/384 = 16$. □

Choose a section $g \mapsto (g, U(g))$ of E^{pair}. Paull's intertwiner for any admissible auxiliary vector is a unit scalar multiple of $U(g)$, so its normalization can be chosen to agree with this section. The cocycle then takes values in μ_2, giving a common linear realization. Comparison with a prescribed normalization on all of $\mathrm{Sp}_4(\mathbb{Z}/4)$ still requires (6.5) and (6.8).

7 Paull's affine comparison theorem and its applications

7.1 The homomorphism to the affine metaplectic group

Let $W = \widetilde{W}/2\widetilde{W}$. Paull constructs a map of base groups

$$\eta : \mathrm{Sp}(\widetilde{W}) \longrightarrow \mathrm{ASp}(B_W).$$

Use the standard polarization

$$B_{\widetilde{W}}((x_1, y_1), (x_2, y_2)) = \langle x_1, y_2 \rangle_{\widetilde{W}}, \qquad B = B_{\mathrm{Weil}},$$

and require the chosen $\widetilde{\chi}$ to be trivial on X. The two choices of w and the affine parameter γ must satisfy the conditions in the original definitions, namely equations (97), (203), (109), and (113) of the uploaded version. Under these assumptions:

1. Lemma 2.71 gives the explicit scalar ratio between the two families of intertwining operators in the mod-4 and affine models. This ratio belongs to μ_4 if and only if
$$\frac{\gamma(\widetilde{\sigma}, \widetilde{w}_{\widetilde{\sigma}})}{\gamma(\sigma)} \in \mu_4, \qquad \sigma = \pi(\widetilde{\sigma}). \tag{7.1}$$
Here π is reduction of the symplectic group modulo 2; η also records the affine correction data.

2. Lemma 2.72 expresses the ratio of the two cocycles as the coboundary of the scalar cochain above.

3. Proposition 2.73 states that, once a μ_2-valued mod-4 normalization has been chosen, there is a homomorphism
$$S : \mathrm{Mp}(\widetilde{W})_{\chi} \longrightarrow \mathrm{AMp}(B_W)_{\widetilde{\chi}} \tag{7.2}$$
compatible with both the projections to the base groups and the specified Weil representations if and only if (7.1) holds for every $\widetilde{\sigma}$. If it exists, it is unique, and
$$S(\widetilde{\sigma}, \pm\omega) = (\eta(\widetilde{\sigma}), \pm\omega), \qquad \ker S = \ker \widetilde{\omega}_{\chi}.$$

The affine extension has kernel μ_4; S is a homomorphism between extensions of different groups, not an isomorphism of double covers.

7.2 The scalar condition in rank at least three

Corollary 7.1. Retain the models, polarization, and central-character conditions above. If $m \geq 3$, then (7.1) holds automatically for every affine normalization satisfying Paull's defining conditions and every valid μ_2-valued mod-4 normalization.

Proof. Write the ratio of the intertwining operators as

$$U_P(g) = a(g) U_{\mathrm{aff}}(\eta(g)).$$

Paull's affine cocycle takes values in μ_4, and the cocycle on the mod-4 side takes values in μ_2. Hence $\partial a \in \mu_4$, so a^4 is a character of $\mathrm{Sp}_{2m}(\mathbb{Z}/4)$. Perfectness forces $a^4 = 1$, and therefore $a(g) \in \mu_4$. Now apply the equivalence in Lemma 2.71 and Proposition 2.73. □

Together with Theorem 6.3, this gives, for all F, n and $M \geq \max(nd, 3)$,

$$S_F : E_{F,n} \longrightarrow \mathrm{AMp}^P_M, \qquad \widetilde{\omega}^P_{\mathrm{aff},M} \circ S_F \simeq \overline{\tau}_L \otimes I_{2^{M-nd}}, \qquad \ker S_F = \ker \overline{\tau}_L. \tag{7.3}$$

The map of base groups is $\eta_M \circ j_M$, and the central μ_2 maps into the sign subgroup of μ_4. Its image is isomorphic to $G_{F,n}$, and need not equal AMp^P_M. When an arbitrary specified affine normalization is used directly at low rank, perfectness is unavailable, and the original phase condition still needs to be checked.

7.3 Relation to earlier local constructions

Paull [4, Section 3.4, pp. 301–308; Proposition 3.55] compares Savin's finite Shimura construction with the two-dimensional model for $F = \mathbb{Q}_2$, $A_1 = \mathrm{SL}_2$. Savin's theorem assumes an unramified field and a simply-laced root system [7]; it does not directly cover arbitrary dyadic F and type C_n. Our comparison instead uses descent modulo four, the trace pairing, and stabilization. Comparison with Paull's pseudosymplectic model also requires compatible subgroup and central-character choices.

8 Finite orthogonal–symplectic duality over $\mathbb{Q}_2$

We study integral images of the classical pair $O(V)\times\mathrm{Sp}(W)$ over $\mathbb{Q}_2$, beginning with the algebraic criterion for their finite multiplicity spaces. Group centralizers will be considered separately in the two-factor example.

8.1 The finite double-commutant criterion

Suppose finite groups K_0, G_0 act linearly and commute on a finite-dimensional complex space $\mathcal{H}$. Write R for the K_0-action and identify $g \in G_0$ with its operator. Put

$$B_0 = \mathbb{C}[K_0], \quad A_0 = \mathbb{C}[G_0], \qquad M_\rho = \mathrm{Hom}_{K_0}(\rho, \mathcal{H}).$$

The brackets here denote the image algebras in $\mathrm{End}_{\mathbb{C}}(\mathcal{H})$. Evaluation gives

$$\mathcal{H} \simeq \bigoplus_{\rho\in\mathrm{Irr}\,K_0} \rho \boxtimes M_\rho.$$

Proposition 8.1 (Double-commutant criterion)**.** The nonzero M_ρ are irreducible and pairwise inequivalent as G_0-modules if and only if $A_0 = B_0'$, equivalently $B_0 = A_0'$. Here primes denote commutants in $\mathrm{End}_{\mathbb{C}}(\mathcal{H})$.

Proof. Semisimplicity and matrix-coefficient orthogonality give

$$B_0 = \bigoplus_{M_\rho\neq 0} \mathrm{End}(\rho)\otimes I, \qquad B_0' = \bigoplus_{M_\rho\neq 0} I\otimes \mathrm{End}(M_\rho).$$

If the multiplicity spaces are irreducible and inequivalent, matrix-coefficient orthogonality for G_0 realizes all the latter matrix blocks independently. Conversely, a reducible multiplicity space prevents a full matrix block, and two equivalent multiplicity spaces force their actions to be linked by an intertwiner. Either possibility contradicts $A_0 = B_0'$. The equivalence with $B_0 = A_0'$ follows from the finite-dimensional semisimple double-commutant theorem. □

Multiplicity-freeness as a $K_0 \times G_0$-module is weaker: in a decomposition $\bigoplus_{\rho,\sigma}(\rho\boxtimes\sigma)^{\oplus m_{\rho\sigma}}$, the criterion requires every row and every column of $(m_{\rho\sigma})$ to have at most one nonzero entry, that entry being one. The isotypic projector and multiplicity character are

$$E_\rho = \frac{\dim\rho}{|K_0|}\sum_{k\in K_0}\overline{\chi_\rho(k)}R(k), \tag{8.1}$$

$$\chi_{M_\rho}(g) = \frac{1}{|K_0|}\sum_{k\in K_0}\overline{\chi_\rho(k)}\,\mathrm{tr}(gR(k)|\mathcal{H}). \tag{8.2}$$

In particular, $\dim E_\rho\mathcal{H} = (\dim\rho)\dim M_\rho$.

8.2 Integral orthogonal reduction

Let $r, s \geq 0$, $t = r + s \geq 1$, and set

$$M = \mathbb{Z}_2^t, \qquad D_{r,s} = \mathrm{diag}(1_r, -1_s), \qquad V = M \otimes_{\mathbb{Z}_2} \mathbb{Q}_2, \qquad b(x,y) = x^T D_{r,s} y.$$

The subscript distinguishes $D_{r,s}$ from the dimension $D = n(2n+1)$ used earlier. On $T = M/2M$, define

$$q_{r,s}(\bar{x}) = x^T D_{r,s} x \pmod 4, \qquad q_{r,s}(a+b) - q_{r,s}(a) - q_{r,s}(b) = 2\beta(a,b), \quad \beta(a,b) = a^T b. \tag{8.3}$$

This is a $\mathbb{Z}/4$-valued quadratic function, not an ordinary $\mathbb{F}_2$-valued quadratic form. Changing a representative by $2z$ does not change it. Its isometries preserve β and hence fix the vector $\mathbf{1} = (1, \ldots, 1)^T$, characterized by $\beta(x, \mathbf{1}) = \beta(x, x)$. In the terminology of [10],

$$\mathrm{St}(T, q_{r,s}) = \{g \in \mathrm{GL}_t(\mathbb{F}_2) : q_{r,s}(gx) = q_{r,s}(x),\ g\mathbf{1} = \mathbf{1}\} = O(T, q_{r,s});$$

"stochastic" here means fixing $\mathbf{1}$, with no probabilistic meaning.

Proposition 8.2 (Surjectivity of orthogonal reduction)**.** Reduction induces a surjection

$$O(M, D_{r,s}) \longrightarrow \mathrm{St}(T, q_{r,s}).$$

Proof. Give M the even bilinear form $2D_{r,s}$. Its integral quadratic form is $Q(x) = x^T D_{r,s} x$, and its discriminant quadratic module is

$$M^{\#}/M = (\tfrac{1}{2}M)/M, \qquad q_M(x/2 + M) = \tfrac{1}{4} x^T D_{r,s} x \pmod{\mathbb{Z}_2}.$$

Its isometry group is $\mathrm{St}(T, q_{r,s})$. The natural map from the isometry group of a nondegenerate quadratic $\mathbb{Z}_2$-lattice to that of its discriminant form is surjective by [9, IV, Corollary 5.9]. Since $O(M, 2D_{r,s}) = O(M, D_{r,s})$, this proves the assertion. □

The factor 2 is essential in this argument: the bilinear unimodular lattice $D_{r,s}$ has trivial discriminant group. An elementary lifting proof, including the initial step modulo 8, appears in Appendix A.

For example, if $(r, s) = (3, 1)$ and $u = \mathbf{1}$, then $u^T D_{r,s} u = 2$, and the integral reflection $h = I - u(u^T D_{r,s})$ reduces to $I + J_t$, where J_t is the all-ones matrix. Thus not every reduced isometry comes from a permutation of coordinates with the same sign. Here $\mathrm{St}(T, q_{3,1}) \simeq S_3 \times C_2$: the vectors with value 3 are e_4 and $\mathbf{1} + e_4$, interchanged by $I + J_t$; the stabilizer of e_4 permutes the first three coordinates.

8.3 The maximal-rank commutant theorem

Let $W = \mathbb{Q}_2^{2n}$, $K_n = \mathrm{Sp}_{2n}(\mathbb{Z}_2)$, and let $\widetilde{K_n}$ be its inverse image in the twofold metaplectic group. With $\psi(1) = i$, write τ_n for the lattice representation on $\mathcal{H}_n = \mathbb{C}[\mathbb{F}_2^n]$ from §4.1. The lattice representation for $V \otimes W$ has space

$$\mathcal{H}_{n,t} = \mathbb{C}[\mathrm{Mat}_{t \times n}(\mathbb{F}_2)] \simeq \mathcal{H}_n^{\otimes r} \otimes \overline{\mathcal{H}_n}^{\otimes s}.$$

The pullback along the symplectic factor is

$$\Delta_{r,s}(g) = \tau_n(g)^{\otimes r} \otimes \overline{\tau_n(g)}^{\otimes s}, \qquad g \in \widetilde{K_n}. \tag{8.4}$$

We also write $\Delta_{r,s}(U) = U^{\otimes r} \otimes \overline{U}^{\otimes s}$ for $U \in \mathrm{Cl}_n$. The sign of a summand of $D_{r,s}$ changes ψ to its conjugate. For $O \in \mathrm{St}(T, q_{r,s})$, the operator $R(O)|X\rangle = |OX\rangle$ commutes with $\Delta_{r,s}$. The

integral orthogonal group acts in the Schrödinger model by changes of variables, up to the scalar of its metaplectic lift [11, II.3–4]:

$$\omega(\widetilde{h})|_{\mathcal{H}_{n,t}} = \lambda(\widetilde{h})R(\bar{h}), \qquad h \in O(M, D_{r,s}).$$

Consequently, if $\mathcal{A}, \mathcal{B}$ are the complex algebras generated by the finite symplectic and orthogonal images, respectively,

$$\mathcal{A} = \mathbb{C}[\Delta_{r,s}(\mathrm{Cl}_n)], \qquad \mathcal{B} = \mathbb{C}[R(\mathrm{St}(T, q_{r,s}))]. \tag{8.5}$$

The first equality uses $\mathrm{Cl}_n = U(1)G_n$; the second uses Proposition 8.2. Scalar factors do not affect these algebras, so no splitting of the orthogonal double cover is assumed.

The rank used next is the rank of a Clifford representation in [10, Section IV]. Its restriction to the real diagonal Clifford subgroup has weights indexed by symmetric bilinear forms on $\mathbb{F}_2^n$; the rank of an irreducible representation is the largest rank of an occurring weight. This is distinct from symplectic rank. Let $\mathcal{H}_{n,t}^{(t)}$ be the sum of the irreducible constituents of rank t, not the span of basis vectors whose labels X have rank t.

Corollary 8.3 (Mutual commutants on the maximal-rank subspace)**.** If $n \geq t$, then on $\mathcal{H}_{n,t}^{(t)}$ the two algebras in (8.5) are mutual commutants. More precisely,

$$\mathcal{H}_{n,t}^{(t)} \simeq \bigoplus_{\rho \in \mathrm{Irr}\,\mathrm{St}(T, q_{r,s})} \rho \boxtimes \eta(\rho),$$

where the $\eta(\rho)$ are pairwise inequivalent irreducible Clifford representations. This formula uses the normalized action R; the labels for the full covering groups must include their central characters.

Proof. Take the isotropic subspace $N = 0$ in [10, Theorem V.1]. Its hypothesis becomes $n \geq t$, and the theorem gives the displayed decomposition with η injective. Semisimplicity then gives the mutual-commutant assertion, and (8.5) identifies these with the algebras of the integral images. □

The case $n = 1, t = 2$ lies outside this hypothesis and is computed below.

8.4 A four-dimensional multiplicity-free example

Take $(r, s, n) = (2, 0, 1)$. Then $q_{2,0}(x) = x_1 + x_2 \pmod 4$ and $\mathrm{St}(T, q_{2,0}) = \langle \mathrm{SWAP} \rangle \simeq C_2$. Use $S, W = \zeta_8 H$ from (4.2) and the ordered basis $e_{00}, e_{01}, e_{10}, e_{11}$ of $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$. Put

$$A_\Delta = S \otimes S = \mathrm{diag}(1, i, i, -1), \qquad B_\Delta = W \otimes W = \frac{i}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{8.6}$$

Let $\Gamma_\Delta = \langle A_\Delta, B_\Delta \rangle$ and $P(e_a \otimes e_b) = e_b \otimes e_a$. Since P commutes with both generators,

$$\mathcal{H}_+ = \mathrm{Sym}^2(\mathbb{C}^2) = \langle u_0, u_1, u_2 \rangle, \qquad \mathcal{H}_- = \Lambda^2(\mathbb{C}^2) = \langle v \rangle,$$

where $u_0 = e_{00}$, $u_1 = (e_{01} + e_{10})/\sqrt{2}$, $u_2 = e_{11}$, and $v = (e_{01} - e_{10})/\sqrt{2}$. The block matrices are

$$A_+ = \mathrm{diag}(1, i, -1), \quad B_+ = \frac{i}{2} \begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}, \qquad A_- = (i), \quad B_- = (-i). \tag{8.7}$$

In particular, the one-dimensional representation is not trivial: $\Lambda^2(U) = \det U$, with $\det S = i$ and $\det W = -i$.

Proposition 8.4. The two blocks are inequivalent irreducible Γ_Δ-modules, and

$$\mathrm{End}_{\Gamma_\Delta}(\mathcal{H}) = \mathbb{C}I \oplus \mathbb{C}P, \qquad \mathbb{C}[\Gamma_\Delta] = \mathrm{End}(\mathcal{H}_+) \oplus \mathrm{End}(\mathcal{H}_-). \tag{8.8}$$

Proof. The distinct eigenvalues of A_+ imply that its invariant subspaces are spans of subsets of u_0, u_1, u_2. The nonzero off-diagonal entries of B_+ connect all three eigenspaces, so no nonzero proper subspace is invariant under both matrices. The other block is one-dimensional; the blocks have different dimensions and are inequivalent. Schur's lemma and semisimplicity give (8.8). □

The projections $E_\pm = (I \pm P)/2$ give the finite theta pairing $\mathrm{Hom}_{\langle P\rangle}(\pm, \mathcal{H}) \simeq \mathcal{H}_\pm$.

8.5 Covering groups and central characters in the example

The integral orthogonal factor must be taken with its complete inverse image in $\mathrm{Mp}_4(\mathbb{Q}_2)$. The Levi scalar has square $(\det h, -1)_{\mathbb{Q}_2}$; thus a coordinate swap has operators $\pm iP$, while $\mathrm{diag}(-1, 1)$, which reduces to the identity, has operators $\pm iI$ [11, I.2, I.4–6]. Since $\det h = \pm 1$, its image is exactly

$$\Gamma_O = \langle iI, P\rangle \simeq C_4 \times C_2.$$

If $\chi_\pm(iI) = i$ and $\chi_\pm(P) = \pm 1$, then as a representation of $\Gamma_\Delta \times \Gamma_O$,

$$\mathcal{H} \simeq \pi_+ \boxtimes \chi_+ \oplus \pi_- \boxtimes \chi_-, \qquad \dim \pi_+ = 3, \quad \dim \pi_- = 1. \tag{8.9}$$

The common element $-I$ has character -1 in each factor, so the product action is well defined on the central product of these groups.

Exact computation with the specified generators gives

$$|\Gamma_\Delta| = 48, \qquad \Gamma_\Delta \simeq \mathrm{SL}_2(\mathbb{Z}/4), \qquad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \mapsto A_\Delta, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mapsto B_\Delta. \tag{8.10}$$

The tensor map $G_1 \to \Gamma_\Delta$ has kernel $\{\pm I_2\}$, since $U \otimes U = I$ implies $U = \pm I_2$. Let $j : \mathrm{Sp}_2(\mathbb{Q}_2) \to \mathrm{Sp}_4(\mathbb{Q}_2)$ be the symplectic-factor embedding, and let ϵ_2, ϵ_4 denote the respective nontrivial covering-kernel elements. Pullback of the full Schrödinger representation along the natural map $\mathrm{Mp}_2 \to \mathrm{Mp}_4$ is the tensor square. On $\mathcal{H}$,

$$(\omega \otimes \omega)(\epsilon_2)|_{\mathcal{H}} = (-I_2) \otimes (-I_2) = I_4.$$

Thus the pullback is not genuine as a representation of Mp_2 and factors through Sp_2 [11, II.3, Corollary 3.3]. In contrast, the full inverse image of the *integral* subgroup $j(K_1)$ in Mp_4 preserves $\mathcal{H}$ and contains ϵ_4, which acts as $-I_4$; its representation on $\mathcal{H}$ is genuine. Its image is still Γ_Δ, since $B_\Delta^2 = -I_4$ is already in that group. The entire rational group does not preserve $\mathcal{H}$, and the finite integral action does not factor through $\mathrm{Sp}_2(\mathbb{F}_2)$, since $A_\Delta^2 = \mathrm{diag}(1, -1, -1, 1) \neq I$.

The quadratic space with Gram matrix $D_{2,0} = I_2$ is anisotropic over $\mathbb{Q}_2$, because -1 is not a square. It cannot be replaced by the hyperbolic space with even integral Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. It is also distinct from the ramified-field lattice representation in Section 4.3, despite the common dimension four.

8.6 Group centralizers and the scope of the Paull comparison

Fix the finite matrix group

$$G_2 = \langle S \otimes I, I \otimes S, W \otimes I, I \otimes W, \mathrm{CNOT}\rangle, \qquad |G_2| = 46080, \quad Z(G_2) = \mu_4 I.$$

Then

$$C_{G_2}(\Gamma_\Delta) = \Gamma_O, \qquad H_{\mathrm{big}} := C_{G_2}(\Gamma_O) = C_{G_2}(P) \supsetneq \mu_4 \Gamma_\Delta. \tag{8.11}$$

Indeed, (8.8) implies that a commuting unitary has the form $U = e^{i\alpha}(cI + isP)$, where $c, s \in \mathbb{R}$ and $c^2 + s^2 = 1$. With the notation $ZI = Z \otimes I$, etc.,

$$U(ZI)U^* = c^2 ZI + s^2 IZ + cs(YX - XY).$$

The four Pauli matrices on the right are linearly independent. Clifford conjugation must give a single signed Pauli matrix, so $cs = 0$. Intersection with G_2 therefore gives precisely $\mu_4\langle P\rangle$. On the other hand, $CZ = \mathrm{diag}(1, 1, 1, -1)$ commutes with P but is not a scalar multiple of any $U \otimes U$: its action on e_{00}, e_{11} would force U diagonal, whereas a scalar tensor square has diagonal entries satisfying $d_{00}d_{11} = d_{01}d_{10}$, which fails for CZ.

Exact closure and centralizer computations give

$$H_{\mathrm{big}} = \langle A_\Delta, B_\Delta, iI, CZ\rangle, \qquad |H_{\mathrm{big}}| = 768, \qquad |\mu_4\Gamma_\Delta| = 96.$$

Since $\Gamma_\Delta \subset H_{\mathrm{big}}$, equation (8.11) implies

$$C_{G_2}(H_{\mathrm{big}}) = \Gamma_O, \qquad C_{G_2}(\Gamma_O) = H_{\mathrm{big}}.$$

The enlarged group preserves both blocks and contains Γ_Δ, so its generated algebra is still $M_3(\mathbb{C}) \oplus \mathbb{C}$. This constructs mutual group centralizers in G_2; the enlarged group is not the direct finite image of the original symplectic factor, and no new classical reductive dual pair is asserted.

The ambient symplectic rank here is $nt = 2$. Transport to a prescribed Paull model requires matching both factor embeddings, lifts, and phases in that dimension, subject to (6.8). The stabilized comparison (6.7) alone cannot preserve this multiplicity-one pairing.

9 Cubic and quartic tensor decompositions

For the remaining tensor calculations the field is $\mathbb{Q}_2$, the powers are pure, and $\psi(1) = i$. Write $d_n = 2^n = \dim \mathcal{H}_n$, distinct from $d = [F : \mathbb{Q}_2]$. The notation $\Theta_{n,t}$ denotes a finite multiplicity space; its local interpretation is proved in Section 12 only for the specified four-dimensional orthogonal space.

9.1 The third and fourth tensor powers for one qubit

For $t = 3, 4$, the only vectors of $\mathbb{F}_2^t$ with $q_{t,0}(x) = \mathrm{wt}(x) \equiv 1 \pmod 4$ are the standard basis vectors. Consequently $\mathrm{St}(\mathbb{F}_2^t, q_{t,0}) = S_t$, acting by tensor permutations. Put

$$V = \mathcal{H}_1 = \mathbb{C}^2, \quad \Gamma_t = \{U^{\otimes t} : U \in G_1\}, \quad \Theta_{1,t}(\rho) = \mathrm{Hom}_{S_t}(\rho, V^{\otimes t}).$$

If a unitary matrix satisfies $U^{\otimes t} = cI$, comparison of eigenvalue products differing in one factor shows that U is scalar. Thus

$$\Gamma_3 \simeq G_1, \quad |\Gamma_3| = 96, \qquad \Gamma_4 \simeq G_1/\mu_4 \simeq S_4, \quad |\Gamma_4| = 24.$$

For the last isomorphism, conjugation on the three Pauli axes identifies G_1/μ_4 with the orientation-preserving signed permutation group in dimension three. Its faithful action on the four body-diagonal lines gives S_4. Label these lines by

$$(1, 1, 1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, -1, 1).$$

Then $S^{\otimes 4}$ corresponds to $(1\,2\,4\,3)$, and $W^{\otimes 4}$ to $(1\,3)$. We use ρ_λ for the orthogonal permutation factor and σ_λ for this second copy of S_4.

The trace of a tensor permutation is $2^{c(\pi)}$, where $c(\pi)$ is its number of cycles. Character inner products give multiplicities $(4, 2, 0)$ for the S_3 types $[3], [21], [111]$, and $(5, 3, 1, 0, 0)$ for the

S_4 types [4], [31], [22], [211], [1111]. More concretely, with $\eta = e_0 \otimes e_1 - e_1 \otimes e_0$, antisymmetrizing the first two factors realizes the cubic [21] multiplicity space as $\langle \eta \rangle \otimes V$. For the fourth power, the images of $(I - R((12)))(I + R((34)))/4$ and $(I - R((12)))(I - R((34)))/4$ realize one copy of the [31] and [22] multiplicity spaces, respectively. Their actions are $\det \otimes \operatorname{Sym}^2 V$ and $\det^2$.

Proposition 9.1 (The two small tensor decompositions)**.** As representations of the normalized orthogonal factor and the actual tensor image,

$$V^{\otimes 3} \simeq \mathbf{1} \boxtimes \alpha_4 \ \oplus \ \rho_{[21]} \boxtimes \alpha_2, \tag{9.1}$$

$$V^{\otimes 4} \simeq \mathbf{1} \boxtimes (\sigma_{[22]} \oplus \sigma_{[211]}) \ \oplus \ \rho_{[31]} \boxtimes \sigma_{[31]} \ \oplus \ \rho_{[22]} \boxtimes \sigma_{[1111]}. \tag{9.2}$$

Here $\alpha_4 = \operatorname{Sym}^3 V$ and $\alpha_2 = \det \otimes V$ are inequivalent irreducible Γ_3-modules of dimensions four and two. All displayed σ's are inequivalent irreducible Γ_4-modules with their indicated partition labels.

Proof. Use the unnormalized symmetric vectors $u_k^{(t)} = \sum_{\mathrm{wt}(x)=k} |x\rangle$. In the ordered cubic symmetric basis, the generators restrict to

$$A_4 = \operatorname{diag}(1, i, -1, -i), \qquad B_4 = \frac{-1+i}{4} \begin{pmatrix} 1 & 3 & 3 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -3 & 3 & -1 \end{pmatrix}.$$

On $\det \otimes V$, they are

$$A_2 = \operatorname{diag}(i, -1), \qquad B_2 = \frac{1-i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The simple spectra of the A's and the nonzero entries of the B's imply irreducibility: spectral projections of A reduce any invariant subspace to a sum of eigenlines, and B connects all of them.

For the fourth power, write $u_k = u_k^{(4)}$. The symmetric space splits as

$$C_2 = \langle u_0 + u_4, u_2 \rangle, \qquad C_3 = \langle u_0 - u_4, u_1, u_3 \rangle.$$

The two generator pairs are

$$A_{C_2} = \operatorname{diag}(1, -1), \quad B_{C_2} = -\frac{1}{2} \begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix},$$

$$A_{C_3} = \operatorname{diag}(1, i, -i), \quad B_{C_3} = \begin{pmatrix} 0 & -1 & -1 \\ -1/2 & -1/2 & 1/2 \\ -1/2 & 1/2 & -1/2 \end{pmatrix}.$$

On $D_3 = \det \otimes \operatorname{Sym}^2 V$ the pair is

$$A_{D_3} = \operatorname{diag}(i, -1, -i), \qquad B_{D_3} = \frac{1}{2} \begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 1 & -2 & 1 \end{pmatrix},$$

and on $D_1 = \det^2$ both generators are -1. Again the simple spectra and the connected nonzero entries prove irreducibility. The traces of A on C_3, D_3, namely $1, -1$, distinguish them. Since A represents a four-cycle, the S_4 character table identifies C_2, C_3, D_3, D_1 with $\sigma_{[22]}, \sigma_{[211]}, \sigma_{[31]}, \sigma_{[1111]}$. The multiplicities and the subspaces obtained by antisymmetrization exhaust the space. □

Thus the cubic decomposition satisfies Proposition 8.1, whereas the fourth power has a reducible multiplicity space for the trivial orthogonal representation. If $A_t = \mathbb{C}[\Gamma_t]$, $B_t = \mathbb{C}[R(S_t)]$, then

t	$\dim A_t$	$\dim B_t$	$\dim A'_t$	$\dim B'_t$
3	20	5	5	20
4	23	14	15	35

These are commutant-algebra dimensions, not group-centralizer orders.

9.2 The Pauli averaging projection

Define, for every $n \geq 1$,

$$P_n^{(4)} = d_n^{-2} \sum_{a,b \in \mathbb{F}_2^n} (X_a Z_b)^{\otimes 4}, \qquad \mathcal{H}_{n,4}^{+} = P_n^{(4)} \mathcal{H}_{n,4}, \quad \mathcal{H}_{n,4}^{-} = (I - P_n^{(4)}) \mathcal{H}_{n,4}. \tag{9.3}$$

Thus $\mathcal{H}_{n,4}^{+} = \operatorname{im} P_n^{(4)}$ and $\mathcal{H}_{n,4}^{-} = \ker P_n^{(4)}$. The signs label these two invariant summands; the eigenvalues of the projection are 1 and 0, not 1 and -1. The fourth tensor power removes the Pauli multiplication phases, so $P_n^{(4)}$ is the orthogonal projection onto the Pauli-invariant subspace. It commutes with both G_n and $R(S_4)$. It is neither zero nor the identity on $\operatorname{Sym}^4 \mathcal{H}_n$: the average of $|0\rangle^{\otimes 4}$ is nonzero, whereas the symmetric sum of labels with one nonzero row b and three zero rows is killed by the average. Indeed, a Z_c with $b \cdot c = 1$ acts by -1 on that sum. Consequently the multiplicity space of the trivial orthogonal representation is reducible for every $n \geq 1$.

For one qubit,

$$P_1^{(4)} = \tfrac{1}{4}(I^{\otimes 4} + X^{\otimes 4} + Y^{\otimes 4} + Z^{\otimes 4}),$$

and its eigenvalues on C_2, C_3, D_3, D_1 are $1, 0, 0, 1$. The first two follow from even weight and invariance under complementing all bits. On D_1 every Pauli determinant has square one; on D_3 there is no simultaneous fixed vector for X, Z. Therefore

$$\begin{aligned} \mathcal{H}_{1,4}^{+} &\simeq \mathbf{1} \boxtimes \sigma_{[22]} \oplus \rho_{[22]} \boxtimes \sigma_{[1111]}, \qquad \dim \mathcal{H}_{1,4}^{+} = 4, \\ \mathcal{H}_{1,4}^{-} &\simeq \mathbf{1} \boxtimes \sigma_{[211]} \oplus \rho_{[31]} \boxtimes \sigma_{[31]}, \qquad \dim \mathcal{H}_{1,4}^{-} = 12. \end{aligned}$$

On each summand the two image algebras are mutual commutants. Adjoining the projector separates the two occurrences of the trivial orthogonal representation.

Theorem 9.2 (Commutants of the fourth tensor power)**.** For every $n \geq 1$, set $A_{n,4} = \mathbb{C}[\{U^{\otimes 4} : U \in G_n\}]$ and $B_{n,4} = \mathbb{C}[R(S_4)]$. Then

$$A'_{n,4} = \langle B_{n,4}, P_n^{(4)} \rangle_{\mathrm{alg}}.$$

On each $\mathcal{H}_{n,4}^{\pm}$, the two image algebras are mutual commutants. All nonzero

$$\Theta_{n,4}^{\pm}(\rho) = \operatorname{Hom}_{S_4}(\rho, \mathcal{H}_{n,4}^{\pm})$$

are irreducible, and all these representations are pairwise inequivalent, including between the two summands.

Proof. The commutant equality is the qubit fourth-tensor commutant theorem [12, Theorem 1 and Proposition 5]. Its projector is exactly (9.3): changing Pauli representatives by μ_4 has no effect, and $\mathrm{Cl}_n = U(1)G_n$ leaves the commutant unchanged. Compressing to either summand makes the projector scalar; every commuting operator on that summand extends by zero on the other. Proposition 8.1 gives the summand pairings. Moreover $P_n^{(4)}$ commutes with every generator of $A'_{n,4}$, so it belongs to $A''_{n,4} = A_{n,4}$. Its distinct eigenvalues on the summands preclude an equivalence between constituents in different summands. □

Taking traces gives

$$\operatorname{rank} P_n^{(4)} = d_n^2, \qquad \operatorname{rank}(P_n^{(4)}|_{\operatorname{Sym}^4 \mathcal{H}_n}) = \frac{(d_n+1)(d_n+2)}{6}.$$

For the second formula, a nontrivial Pauli representative has eigenvalues ± 1 or $\pm i$, each with multiplicity $d_n/2$. Its symmetric-fourth-power trace is $d_n(d_n+2)/8$. Averaging this with the identity contribution $\binom{d_n+3}{4}$ proves the formula.

10 The fourth-tensor code and its irreducible constituents

In this section assume $n \geq 4$, and keep $d_n = 2^n$. Write tensor labels as $X \in \mathrm{Mat}_{4\times n}(\mathbb{F}_2)$, with four rows for the tensor factors. Set

$$T = \mathbb{F}_2^4, \quad q(x) = \mathrm{wt}(x) \pmod 4, \quad \beta(x,y) = x^T y, \quad N = \langle \mathbf{1} \rangle, \quad \mathbf{1} = (1,1,1,1)^T.$$

The only subspaces on which q vanishes are $0, N$, and both are stochastic in the terminology of [10]: $\mathbf{1} \in N^\perp$. Both stabilizers in $\mathrm{St}(T,q) = S_4$ are all of S_4.

10.1 The code and its quotient action

For columns of X in $N^\perp$, define

$$|[X]_N\rangle = d_n^{-1/2} \sum_{a \in \mathbb{F}_2^n} |X + \mathbf{1}a^T\rangle.$$

One vector for each $\mathrm{Hom}(\mathbb{F}_2^n, N^\perp/N)$ label gives an orthonormal basis. Acting on a computational basis vector yields

$$P_n^{(4)}|X\rangle = \begin{cases} d_n^{-1} \sum_a |X + \mathbf{1}a^T\rangle, & X^T\mathbf{1} = 0, \\ 0, & X^T\mathbf{1} \neq 0. \end{cases} \tag{10.1}$$

Thus the Pauli-invariant summand is the code space

$$C_N = \mathrm{CSS}(N)^{\otimes n} = \mathcal{H}_{n,4}^+, \qquad \dim C_N = 4^n = d_n^2.$$

Here $\mathrm{CSS}(N)$ is the Calderbank–Shor–Steane code spanned by $(|u\rangle + |u+\mathbf{1}\rangle)/\sqrt{2}$, for $u \in N^\perp$.

In $T_N = N^\perp/N$, take $\bar{u} = [1100]$, $\bar{v} = [1010]$. Then

$$q_N(a\bar{u} + b\bar{v}) = 2(a + b + ab) \pmod 4.$$

Every nonzero vector has value two, and $\beta_N(\bar{u}, \bar{v}) = 1$. Hence $\mathrm{St}(T_N, q_N) = \mathrm{GL}_2(\mathbb{F}_2) \simeq S_3$. The actual quotient map

$$p : S_4 \longrightarrow S_3$$

permutes $12|34, 13|24, 14|23$; its kernel is the Klein four group V_4. It acts faithfully on the code after this quotient. For the three S_3-types,

$$p^*\tau_{[3]} = \rho_{[4]}, \qquad p^*\tau_{[21]} = \rho_{[22]}, \qquad p^*\tau_{[111]} = \rho_{[1111]}.$$

10.2 Clifford representation rank

The normalized real diagonal Clifford subgroup has operators

$$U_f|x\rangle = (-1)^{f(x)}|x\rangle, \qquad f(x) = \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j.$$

Its characters are indexed by symmetric matrices B, with exponent $\sum_i a_i B_{ii} + \sum_{i<j} b_{ij} B_{ij}$. Direct calculation gives weight matrix $X^T X$ on $|X\rangle$. The rank of a representation is the maximum rank of its occurring weight matrices, as in [10, Section IV]. For a code vector the weight is unchanged by $X \mapsto X + \mathbf{1}a^T$ and is the pullback of β_N, so its rank is at most two. Every weight in the full tensor space has rank at most four.

These bounds are attained in every permitted multiplicity space. For the $-$-summand, $X_- = (I_4\ 0)$ has a free S_4 orbit of 24 basis vectors, all killed by $P_n^{(4)}$, and all of weight $\mathrm{diag}(I_4, 0)$. Each S_4-type occurs in this regular orbit. For the $+$-summand, take first columns $1100, 1010$ and remaining columns zero. Its code label has a free S_3 orbit of six vectors, all of weight $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \oplus 0$. Every quotient type occurs. More explicitly, the squared norm of the isotypic projection of a regular-orbit basis vector is $(\dim \rho)^2/24$, or $(\dim \tau)^2/6$, respectively. The projection preserves the common weight space. Together with Theorem 9.2, this proves that the $+$-summand consists exactly of rank-two constituents and the $-$-summand exactly of rank-four constituents.

The finite Clifford group used in [10] satisfies

$$\mathrm{Cl}_n^{\mathrm{MG}} = \mu_8 G_n, \qquad \Delta_{4,0}(\zeta_8 I) = -I.$$

Indeed it uses H_j, S_j, CNOT, while our Weyl operator is $W_j = \zeta_8 H_j$; also $(HS)^3 = \zeta_8 I$. All fourth-power constituents have this same extra scalar action. Restriction to G_n therefore preserves their irreducibility and inequivalence. The real diagonal subgroup itself lies in G_n, so representation rank is preserved as well.

Theorem V.1 of [10] applies to an isotropic subspace of dimension m when $m \geq (4-n)/2$. For 0 it requires $n \geq 4$, and for N it requires $n \geq 2$. In our common range it consequently gives

$$\Theta^-_{n,4}(\rho_\lambda) \simeq \mathrm{Res}_{G_n}\, \eta_0(\rho_\lambda), \qquad \lambda \vdash 4, \tag{10.2}$$

$$\Theta^+_{n,4}(p^*\tau) \simeq \mathrm{Res}_{G_n}\, \eta_N(\tau), \qquad \tau \in \mathrm{Irr}\, S_3. \tag{10.3}$$

These are the multiplicity factors of the same joint actions on the spaces just identified. The rank-four endpoint $n = 4$ is covered by the regular-orbit argument; it does not use the stricter inequality in Theorem V.14 of that source.

10.3 Multiplicity dimensions and isotypic projections

On the five conjugacy classes $1^4, 21^2, 2^2, 31, 4$, the full tensor character is $(d_n^4, d_n^3, d_n^2, d_n^2, d_n)$. The character on the code is $(d_n^2, d_n, d_n^2, 1, d_n)$: a quotient permutation fixes respectively $4, 2, 4, 1, 2$ of the four vectors of T_N, independently in each column. The character on the $-$-summand is their difference. Taking S_4 character inner products gives the complete table below. Entries are multiplicity space dimensions, to be multiplied by $\dim \rho_\lambda$ when computing isotypic dimensions.

Summand	λ	Rank	$\dim \Theta^{\pm}_{n,4}(\rho_\lambda)$	$n = 4$	$\dim \rho_\lambda$
$+$	[4]	2	$(d_n+1)(d_n+2)/6$	51	1
$+$	[22]	2	$(d_n^2-1)/3$	85	2
$+$	[1111]	2	$(d_n-1)(d_n-2)/6$	35	1
$-$	[4]	4	$(d_n-1)(d_n+1)(d_n+2)(d_n+4)/24$	3825	1
$-$	[31]	4	$d_n(d_n+2)(d_n^2-1)/8$	9180	3
$-$	[22]	4	$(d_n^2-4)(d_n^2-1)/12$	5355	2
$-$	[211]	4	$d_n(d_n-2)(d_n^2-1)/8$	7140	3
$-$	[1111]	4	$(d_n+1)(d_n-1)(d_n-2)(d_n-4)/24$	1785	1

The positive [31] and [211] multiplicities vanish. The dimension sums, weighted by the final column, are d_n^2 and $d_n^4 - d_n^2$, respectively. At $n = 4$ these are 256 and 65280. The three quotient labels occur in both summands, so the eight-term jointly multiplicity-free decomposition is not a single bijection between orthogonal and symplectic types on the whole space.

With E_{ρ_λ} as in (8.1), the joint isotypic projectors are $P_n^{(4)} E_{\rho_\lambda}$ and $(I - P_n^{(4)}) E_{\rho_\lambda}$. If D_λ is a unitary model of ρ_λ, the image of

$$P^{\pm} \frac{\dim \rho_\lambda}{24} \sum_{\pi \in S_4} \overline{D_\lambda(\pi)_{aa}} R(\pi), \qquad P^+ = P_n^{(4)}, \quad P^- = I - P_n^{(4)},$$

is one concrete copy of the corresponding multiplicity space. Its character on $U \in G_n$ is

$$\frac{1}{24} \sum_{\pi \in S_4} \overline{\chi_\lambda(\pi)} \operatorname{tr}(U^{\otimes 4} R(\pi) P^{\pm}).$$

For the full tensor trace the summand is $\prod_c \operatorname{tr}(U^{|c|})$, over cycles c of π; for the positive trace it is $d_n^{-2} \sum_{a,b} \prod_c \operatorname{tr}((U X_a Z_b)^{|c|})$. These formulas retain the operator phases.

Finally, the three positive G_n-representations descend to $G_n / P_n \simeq \mathrm{Sp}_{2n}(\mathbb{F}_2)$, whereas the $-$-summand has no Pauli-fixed vectors. For $\mathrm{Cl}_n^{\mathrm{MG}}$, the additional $\zeta_8 I$ still acts as $-I$ and cannot be discarded in this descent.

11 Metaplectic inverse images and the fourth tensor power

Keep $n \geq 4$, $M = (\mathbb{Z}_2^4, I_4)$, $W_n = \mathbb{Q}_2^{2n}$, and $\mathscr{W} = M_F \otimes W_n$. The base embeddings are

$$j_4(g) = I_4 \otimes g, \qquad \iota_n(h) = h \otimes I_{W_n}.$$

Their full integral inverse images in the twofold $p_{\mathscr{W}} : \mathrm{Mp}(\mathscr{W}) \to \mathrm{Sp}(\mathscr{W})$ are denoted by

$$\widetilde{K}_{n,4}^{\mathscr{W}} = p_{\mathscr{W}}^{-1}(j_4(K_n)), \qquad \widetilde{O}_{n,4} = p_{\mathscr{W}}^{-1}(\iota_n(O(M))).$$

Both preserve $\mathcal{H}_{n,4} = \mathcal{S}(M^n/(2M)^n)$.

11.1 The tensor splitting and the symplectic image

Proposition 11.1 (Splitting induced by the fourth tensor power)**.** There is a homomorphism

$$J_4 : \mathrm{Mp}(W_n) \longrightarrow \mathrm{Mp}(\mathscr{W}), \qquad \ker J_4 = \langle \epsilon_n \rangle,$$

whose Weil operator is the fourth tensor power of the source operator. It induces an injective homomorphism $s_4 : \mathrm{Sp}(W_n) \to \mathrm{Mp}(\mathscr{W})$, with $p_{\mathscr{W}} s_4 = j_4$, and

$$p_{\mathscr{W}}^{-1}(j_4(\mathrm{Sp}(W_n))) = \langle \epsilon_{\mathscr{W}} \rangle \times s_4(\mathrm{Sp}(W_n)).$$

Proof. Let $\mathcal{E}(W_n)$ denote the $\mathbb{C}^\times$ extension of Heisenberg intertwiners, as in [11, I.1–2]. Orthogonal direct sum gives the actual homomorphism $(g, T) \mapsto (j_4(g), T^{\otimes 4})$, whose central map is $z \mapsto z^4$. To check its restriction to the double cover, use the characters $\Lambda_n, \Lambda_{\mathscr{W}}$ of [11, I.4–5], with central restriction $z \mapsto z^2$ and kernels the double covers. The quotient of $\Lambda_{\mathscr{W}} \circ \mathcal{J}_4$ and Λ_n^4 is trivial on the center, hence descends to a character of the perfect group $\mathrm{Sp}(W_n)$. It is therefore trivial. The tensor map restricts to the double covers and has kernel precisely the source μ_2. The induced section has trivial intersection with the ambient kernel, and every base element has exactly two lifts. This proves the assertions. The source finite lattice space is stable, so the tensor operator and both ambient lifts preserve $\mathcal{H}_{n,4}$. □

Set $\Gamma_{n,4} = \{U^{\otimes 4} : U \in G_n\}$, and let $\widehat{\Gamma}_{n,4}$ be the image of $\widetilde{K}^{\mathscr{W}}_{n,4}$. The scalar tensor criterion used above proves

$$\Gamma_{n,4} \simeq G_n/\mu_4, \qquad \Gamma_{n,4} \cap \mathbb{C}^\times I = \{I\}, \qquad \widehat{\Gamma}_{n,4} = \mu_2 I \times \Gamma_{n,4}. \tag{11.1}$$

Its center is $Z(\Gamma_{n,4}) = 1$, hence $Z(\widehat{\Gamma}_{n,4}) = \mu_2 I$. Indeed the normal subgroup $P_n/\mu_4 \simeq \mathbb{F}_2^{2n}$ is acted on through the faithful standard symplectic action. A central element must have trivial symplectic image, and its remaining Pauli label must be fixed by every symplectic transformation. No nonzero label has this property, as a suitable transvection moves it.

Define

$$\mathcal{N}_n = \ker \tau_n, \qquad \mathcal{C}_n = \tau_n^{-1}(\mu_4 I), \qquad \mathcal{D}_n = p_n(\mathcal{C}_n) \subset K_n.$$

Then $\ker \tau_n^{\otimes 4} = \mathcal{C}_n$, $\mathcal{C}_n/\mathcal{N}_n \simeq \mu_4$, and the kernel of the action of the full integral preimage is $s_4(\mathcal{D}_n)$. The source kernel ϵ_n belongs to $\mathcal{C}_n$, but the ambient kernel $\epsilon_{\mathscr{W}}$ acts as $-I$ and is not in the ambient representation kernel. In the mod-4 notation of §4.1, $\mathcal{D}_n$ is the inverse image of the subgroup

$$I + 2\begin{pmatrix} a & b \\ c & a^T \end{pmatrix}, \qquad b = b^T, \quad c = c^T, \quad \operatorname{diag} b = \operatorname{diag} c = 0;$$

the additional condition $\operatorname{tr} a = 0$ is needed for the projection of $\mathcal{N}_n$, not for $\mathcal{D}_n$.

11.2 The orthogonal cover and its parity dependence

Proposition 11.2 (The complete orthogonal image)**.** Put $c_n = 2$ for even n, and $c_n = 4$ for odd n. There is an isomorphism of covering groups

$$\widetilde{O}_{n,4} \simeq \{(h, z) \in O(M) \times \mu_{c_n} : z^2 = (\det h)^n\}, \tag{11.2}$$

with componentwise multiplication. Its finite image and kernel are

$$\Gamma_{O,n,4} = \mu_{c_n} I \times R(S_4), \qquad Z(\Gamma_{O,n,4}) = \mu_{c_n} I, \tag{11.3}$$

$$\ker(\widetilde{O}_{n,4} \to \Gamma_{O,n,4}) = \{(h, 1) : \bar{h} = 1,\ (\det h)^n = 1\}. \tag{11.4}$$

The operator of (h, z) on the finite space is $zR(\bar{h})$.

Proof. On the full Schwartz–Bruhat space set $(T_h f)(Z) = f(h^{-1}Z)$. The corresponding Levi matrix is $A_h = h \otimes I_n$, of determinant $(\det h)^n$. The two lift operators are $\lambda(\widetilde{h})T_h$ with opposite scalars. The normalized Weil-index formula in [11, I.4, Lemma 4.1; I.5] gives

$$\lambda(\widetilde{h})^2 = ((\det h)^n, -1)_{\mathbb{Q}_2} = (\det h)^n,$$

since $\det h = \pm 1$ and $(-1, -1)_{\mathbb{Q}_2} = -1$. In that source's half-cocycle convention use $\psi_K(a) = \psi(2a)$, so the Weil-index character is again ψ. Because $T_{h_1} T_{h_2} = T_{h_1 h_2}$, the scalar is a character on the full inverse image. The two scalars in each fiber give exactly (11.2).

The reduction of $O(M)$ is S_4, since it preserves the weight-mod-4 function and contains all coordinate permutations. The reflection $r = \operatorname{diag}(-1, 1, 1, 1)$ reduces to the identity and has lift scalars ± 1 for even n, and $\pm i$ for odd n. Thus all μ_{c_n} scalars occur. Combining them with lifts of coordinate permutations realizes every pure $R(\pi)$. For odd n, a pure odd permutation can be lifted above rh_π, not above the same odd permutation matrix h_π. The intersection of scalars with $R(S_4)$ is trivial, since these operators fix the zero label and act faithfully on standard basis columns. This also proves the kernel formula and the center assertion. □

The complete orthogonal cover splits for even n. For odd n, each lift of r squares to $\epsilon_{\mathscr{W}}$, so it does not split. For the precise root-word lift obtained by taking the $\widetilde{h}(-1)$ of §4.1 in each of the n planes affected by r, the finite operator is

$$\omega(\widetilde{r}_n)|_{\mathcal{H}_{n,4}} = (-i)^n I.$$

At $n = 4, 5$ this is respectively $I, -iI$; the other lift has the opposite sign. These phases belong to the complete orthogonal image, not to the source scalars of the fourth tensor power.

11.3 The joint image and genuine representations

The groups $\widehat{\Gamma}_{n,4}$ and $\Gamma_{O,n,4}$ commute. An element of their intersection is central in the first, so

$$\widehat{\Gamma}_{n,4} \cap \Gamma_{O,n,4} = \mu_2 I.$$

Their product group is consequently

$$\mathcal{J}_{n,4} \simeq \frac{\widehat{\Gamma}_{n,4} \times \Gamma_{O,n,4}}{\{(zI, z^{-1}I) : z \in \mu_2\}} \simeq \mu_{c_n} \times \Gamma_{n,4} \times S_4.$$

By contrast, the two full integral inverse images intersect in four elements above $\{\pm I_{\mathscr{W}}\}$. The nonidentity element $s_4(-I_{W_n})$ acts trivially on the finite space; thus the group intersection and its finite operator image have different orders.

Let $\chi_c(z) = z$. Extend every finite multiplicity representation by

$$\widehat{\Theta}^{\pm}_{n,4}(\rho)(\varepsilon\gamma) = \varepsilon\Theta^{\pm}_{n,4}(\rho)(\gamma), \qquad \rho^{\mathrm{gen}}_{c_n}(zR(\pi)) = z\rho(\pi).$$

On $\widetilde{O}_{n,4}$ the latter is $\lambda(\widetilde{h})\rho(\bar{h})$. The full joint decomposition, in symplectic–orthogonal order, is

$$\begin{aligned}\mathcal{H}_{n,4} \simeq \bigoplus_{\lambda\in\{[4],[22],[1111]\}} \widehat{\Theta}^{+}_{n,4}(\rho_\lambda) \boxtimes (\rho_\lambda)^{\mathrm{gen}}_{c_n} \\ \oplus \bigoplus_{\lambda\vdash 4} \widehat{\Theta}^{-}_{n,4}(\rho_\lambda) \boxtimes (\rho_\lambda)^{\mathrm{gen}}_{c_n}.\end{aligned} \tag{11.5}$$

Each factor is genuine, and the anti-diagonal common center acts trivially on its external tensor product. The eight symplectic factors remain inequivalent and their dimensions are unchanged. The three repeated orthogonal labels remain repeated. For odd n the actual central character is χ_4, not χ_4^{-1}, although both are genuine. Genuineness alone does not specify the action of iI.

For comparison, at $n = 1, t = 3, 4$ the complete symplectic images are $\widehat{\Gamma}_3 = \Gamma_3$ and $\widehat{\Gamma}_4 = \mu_2 \times \Gamma_4$; the complete orthogonal images are $\mu_4 \times S_3$ and $\mu_4 \times S_4$. Indeed the tensor pullback has source central action $(-1)^t$, whereas the ambient kernel always acts by -1; a determinant-minus-one orthogonal reflection supplies $\pm iI$. Their intersections are $\mu_4 I$ and $\mu_2 I$, respectively: an equality $U^{\otimes t} = cR(\pi)$ first forces U to be diagonal by the all-zero and all-one labels, and the weight-one labels then force $\pi = 1$. Hence the small decompositions above acquire the same actual scalar characters, without changing their invariant subspaces or commutants.

12 An explicit comparison with local theta over $\mathbb{Q}_2$

We now fix $n = 4$ and $M = (\mathbb{Z}_2^4, I_4)$. Put

$$\Omega = \mathcal{S}(\mathrm{Mat}_{4\times 4}(F)), \qquad \mathcal{H} = \mathcal{S}(M^4/(2M)^4), \qquad F = \mathbb{Q}_2.$$

Here L^4 denotes matrices whose four columns lie in a lattice L. Let

$$H_0 = O(M_F), \quad K_0 = O(M), \quad \widetilde{H} = p_{\mathscr{W}}^{-1}(\iota_4(H_0)), \quad \widetilde{G} = p_{\mathscr{W}}^{-1}(j_4(\mathrm{Sp}_8(F))),$$

and let $\widetilde{K_O}, \widetilde{K_G}$ be their full integral inverse images. The integral orthogonal type of interest is $\kappa = \chi_2 \boxtimes \rho_{[4]}$, inflated to $\widetilde{K_O}$. Evaluation on its one-dimensional model identifies

$$F_\kappa = \mathrm{Hom}_{\widetilde{K_O}}(\kappa, \mathcal{H}) = \mathcal{H}^{R(S_4)} = F^+ \oplus F^-, \qquad \dim F^+ = 51, \quad \dim F^- = 3825.$$

These are the two inequivalent irreducible $\widetilde{K_G}$-modules from (11.5) with trivial permutation label. All local Hom spaces below are taken in complex smooth representations.

12.1 The compact orthogonal group and its three lattice translates

Lemma 12.1. The quadratic space M_F is anisotropic, H_0 is compact, and K_0 is open of index three. The orbit of M consists of the three lattices described below.

Proof. Set $Q(x) = x^T x$ and $B(x, y) = x^T y$. If $u \in M \setminus 2M$ has k odd coordinates, reduction of squares modulo eight gives

$$v_2(Q(u)) = \begin{cases} 0, & k = 1, 3, \\ 1, & k = 2, \\ 2, & k = 4. \end{cases}$$

Writing any nonzero vector as $2^a u$ proves anisotropy. It also shows that $Q(x) = 1$ implies $x \in \frac{1}{2}M$. Thus H_0 is a closed subset of the compact set $\frac{1}{2}\operatorname{Mat}_4(\mathbb{Z}_2)$, hence compact; its integral subgroup is open.

Within this section, D denotes the lattice

$$D = \{x \in M : \textstyle\sum_i x_i \in 2\mathbb{Z}_2\} = \{x \in F^4 : Q(x) \in 2\mathbb{Z}_2\}, \qquad D^* = M + \mathbb{Z}_2 v, \quad v = \tfrac{1}{2}(1, 1, 1, 1)^T,$$

not the dimension $n(2n + 1)$ of §1.3. The valuation formula proves the intrinsic expression for D. The lattice D^* is its dual for $B = I_4$: pairing with $2e_i, e_i - e_j \in D$ forces coordinates to be all integral or all half-integral, and the converse follows directly. Hence both lattices are H_0-invariant. We have $[M : D] = [D^* : M] = 2$, $2D^* \subset D$, and $D^*/D \simeq \mathbb{F}_2^2$. Every hM is an index-two intermediate lattice, so it is one of

$$M_0 = M, \qquad M_+ = D + \mathbb{Z}_2 v, \qquad M_- = D + \mathbb{Z}_2(v + e_1).$$

All three occur. Indeed, take

$$t = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad r = \operatorname{diag}(-1, 1, 1, 1). \tag{12.1}$$

Then $t^T = t$, $t^2 = I$, $tM = M_+$, and $rtM = M_-$. The stabilizer of M is K_0, proving

$$H_0 = K_0 \sqcup tK_0 \sqcup rtK_0. \tag{12.2}$$

Distinct lattices satisfy $M_i \cap M_j = D$ and $M_i + M_j = D^*$, as their images are distinct lines in D^*/D. □

The cosets in (12.2) are left cosets; normality of K_0 is not required.

12.2 The rational orthogonal character and averaging map

For every $h \in H_0$, the rational Levi matrix $A_h = h \otimes I_4$ has determinant one. The full Levi formula therefore gives

$$\omega(\widetilde{h}) = \sigma_0(\widetilde{h})T_h, \qquad (T_h f)(Z) = f(h^{-1}Z), \qquad \sigma_0(\widetilde{h}) \in \{\pm 1\}.$$

The scalar ratio is a character because T is a representation. It is genuine, restricts to κ, and is smooth: its kernel on the open integral subgroup is open. The unique lift with scalar one defines $s_O : H_0 \to \widetilde{H}$, giving

$$\widetilde{H} = \langle \epsilon_{\mathscr{W}} \rangle \times s_O(H_0), \qquad \sigma_0(\epsilon_{\mathscr{W}}^a s_O(h)) = (-1)^a.$$

The two full rational inverse images commute. To see this at the group level, fix $\widetilde{h} \in \widetilde{H}$. The map $g \mapsto [\widetilde{h}, s_4(g)]$ is a character of $\mathrm{Sp}_8(F)$ into the central μ_2, hence trivial by perfectness.

Their intersection is $\{1, \epsilon_{\mathscr{W}}, c, \epsilon_{\mathscr{W}} c\}$, where $c = s_4(-I_{W_4}) = s_O(-I_M)$. On Ω, c is the parity operator $f(Z) \mapsto f(-Z)$, whereas on $\mathcal{H}$ it is the identity. We have $\sigma_0(c) = 1$.

Evaluate Hom maps at 1 in the model of σ_0, so that

$$\mathcal{M}(\sigma_0) := \mathrm{Hom}_{\widetilde{H}}(\sigma_0, \Omega) = \Omega^{T(H_0)}, \qquad \mathrm{Hom}_{\widetilde{K}_O}(\kappa, \Omega) = \Omega^{T(K_0)}.$$

Normalize Haar measure on H_0 to mass one. The operator

$$e_{\sigma_0} f = \int_{\widetilde{H}} \sigma_0(\widetilde{h})^{-1} \omega(\widetilde{h}) f \, d\widetilde{h} = \int_{H_0} T_h f \, dh$$

is an algebraic finite sum on each smooth vector, because an open stabilizer has finite index in this compact group. It is the orthogonal isotypic projection and commutes with $\widetilde{G}$. Restricting it gives a $\widetilde{K}_G$-equivariant map

$$\mathcal{A} : F_\kappa \longrightarrow \mathcal{M}(\sigma_0), \qquad \mathcal{A}f = \tfrac{1}{3}(f + T_t f + T_{rt} f). \tag{12.3}$$

Pointwise this is $(f(Z) + f(tZ) + f(trZ))/3$, since $(rt)^{-1} = tr$. The average is taken in Ω, without projecting back to $\mathcal{H}$. Its support is contained in $M_0^4 \cup M_+^4 \cup M_-^4$, and it is constant modulo $(2D)^4$. In each member of that union all columns belong to the same lattice.

12.3 The compression identity and exact images

Write $P = P_4^{(4)}$. Formula (10.1) becomes

$$(Pf)(Z) = \mathbf{1}_{D^4}(Z) \frac{1}{16} \sum_{a \in \mathbb{F}_2^4} f(Z + \mathbf{1}a^T), \qquad P\mathcal{H} = C_N = \mathcal{S}(D^4/(2D^*)^4).$$

Here a uses zero-one representatives; $2M$-periodicity makes the choice immaterial. For the finite delta vector $\delta_0 = \mathbf{1}_{(2M)^4}$, we have

$$f_+ = P\delta_0 = \tfrac{1}{16} \mathbf{1}_{(2D^*)^4} \in F^+. \tag{12.4}$$

The intrinsic lattice D^* makes this vector H_0-invariant. Its $\widetilde{K}_G$-span is all of the irreducible F^+, and the complete actions commute. Hence every vector in F^+ is H_0-invariant, and $\mathcal{A}|_{F^+} = I$.

Take $\mathrm{vol}(M^4) = 1$, and denote by $E_{\mathcal{H}}$ the $L^2(F^{16})$ orthogonal projection onto $\mathcal{H}$:

$$(E_{\mathcal{H}} u)(Z) = \mathbf{1}_{M^4}(Z) \frac{1}{\mathrm{vol}((2M)^4)} \int_{(2M)^4} u(Z + Y) \, dY.$$

Theorem 12.2 (Exact finite-to-local comparison)**.** On the finite-dimensional source F_κ,

$$E_{\mathcal{H}} \mathcal{A} = \mathcal{A}^* \mathcal{A} = P + \tfrac{1}{3}(I - P). \tag{12.5}$$

Both restrictions $\mathcal{A}|_{F^\pm}$ are injective, with orthogonal images of dimensions 51 and 3825. Furthermore,

$$\mathcal{A}(F^+) = F^+, \qquad \mathcal{A}(F^-) \cap \mathcal{H} = 0, \qquad \|\mathcal{A} f_-\|^2 = \tfrac{1}{3}\|f_-\|^2 \quad (f_- \in F^-).$$

The total image has dimension 3876, and its inverse is explicitly $f = (P + 3(I - P)) E_{\mathcal{H}} \mathcal{A} f$.

Proof. For $h = t, rt$, the fundamental compression identity is

$$E_{\mathcal{H}} T_h |_{\mathcal{H}} = T_h P.$$

For one column, $h^{-1}(2M) \subset M$ and $M \cap hM = D$. The compression therefore vanishes outside D. Within D, the image of $h^{-1}(2M)$ modulo $2M$ consists of $\{0, \mathbf{1}\}$, with uniform Haar weights.

Four columns give the 16 equally weighted translates in the formula for P. Since $hD = D$, the identity also holds outside M^4, where both sides vanish. For $f \in F_\kappa$, the vector $Pf \in F^+$ is fixed by both T_t, T_{rt}. Applying the compression to (12.3) proves the first equality with $P + (I-P)/3$.

Since e_{σ_0} is a self-adjoint projection, $\langle \mathcal{A}f, \mathcal{A}g\rangle = \langle f, E_{\mathcal{H}}\mathcal{A}g\rangle$. This proves the adjoint identity and injectivity, orthogonality, and the norm factors. If $w = \mathcal{A}f_-$ lies in $\mathcal{H}$, compression gives $w = f_-/3$; averaging again gives $w = w/3$, so $w = 0$. The inverse formula follows by inverting the two nonzero eigenvalues. □

The adjoint in (12.5) has a finite-dimensional source and the inner product induced from L^2; it does not identify the entire smooth multiplicity space with a finite-dimensional Hilbert space.

There is also a pointwise illustration of the negative image. Put $f_- = (I-P)\delta_0$. Then

$$\mathcal{A}f_- = \tfrac{1}{3}(\mathbf{1}_{(2M_0)^4} + \mathbf{1}_{(2M_+)^4} + \mathbf{1}_{(2M_-)^4}) - \tfrac{1}{16}\mathbf{1}_{(2D^*)^4}. \tag{12.6}$$

In $(2D^*/2D)^4$, classify the four column classes by the dimension of their span. At dimensions zero, one, two the values are, respectively, $15/16, 13/48, -1/16$, occurring on $1, 45, 210$ classes. Hence the support is exactly $(2D^*)^4 \subset M^4$. Nevertheless the value at zero is $15/16$, while the matrix with first column $2e_1$ and other columns zero has value $13/48$. These matrices differ by an element of $(2M)^4$, proving directly that the function is not in $\mathcal{H}$. Its support is integral, but it is not invariant under translation by $(2M)^4$.

12.4 Identification with one irreducible local theta lift

Theorem 12.3. For $\psi_K(a) = \psi(2a)$, restriction along the specified s_4 gives

$$\Theta_0 := \mathcal{M}(\sigma_0)|_{s_4(\mathrm{Sp}_8(F))} \simeq \Theta_{\psi_K}(\mathbf{1}_{H_0}) = \theta_{\psi_K}(\mathbf{1}_{H_0}) \neq 0,$$

and this representation is irreducible. On the full inverse image $\widetilde{G}$, $\mathcal{M}(\sigma_0)$ is genuine and irreducible. Each of $\mathcal{A}(F^+)$ and $\mathcal{A}(F^-)$ generates it.

Proof. A smooth vector for the compact group $\widetilde{H}$ generates a finite-dimensional representation factoring through a finite quotient. For the ordinary action, the averaging operator satisfies

$$\ker e_H = \mathrm{span}_{\mathbb{C}}\{T_h f - f : h \in H_0,\ f \in \Omega\}.$$

Every displayed difference is killed by averaging, and $f - e_H f$ is a finite average of such differences. Hence invariants identify with coinvariants. After the genuine scalar character is removed, the averaging projection consequently identifies the maximal σ_0-isotypic quotient with

$$\Omega(\sigma_0) = \Omega/\ker e_{\sigma_0} \simeq \sigma_0 \boxtimes \mathcal{M}(\sigma_0), \qquad \mathcal{M}(\sigma_0) \simeq (\Omega \otimes \sigma_0^\vee)_{\widetilde{H}}.$$

This is the big theta quotient. On $s_O(H_0)$, σ_0 becomes the trivial representation, and T_h is the traditional orthogonal splitting of [11, II.4]. The symplectic splitting also agrees: the even-dimensional construction in that source twists the intertwiner tensor action by Λ^{-2}, which is one on the source double cover. The half-cocycle character is ψ_K, as stipulated above.

The trivial representation of the full group $H_0 = O(M_F)$ is irreducible, admissible, and supercuspidal, since the group is compact and its quadratic space is anisotropic. The supercuspidal-input theorem recorded in [13, Theorem 1.1(i)] says that its big theta lift is zero or irreducible and equals its small theta lift. That clause does not exclude residue characteristic two; the groups there are the full isometry groups, including this orthogonal group. Both spaces here have even dimension, so the partner is the ordinary $\mathrm{Sp}_8(F)$ with the chosen splitting. Non-vanishing follows already from (12.4): its H_0-invariance follows from the intrinsic lattice D^*, independently of the irreducibility or dimensions of the finite summands. The ambient kernel adds the scalar -1, preserving irreducibility and making the full-cover action genuine. Each nonzero finite-type image generates the irreducible space. □

The dimension $3876 = 51 + 3825$ is that of $\mathcal{A}(F_\kappa)$, not of Θ_0. The construction does not determine the full restriction of Θ_0 to $\widetilde{K}_G$ or the multiplicities in that restriction.

12.5 Embedding into a degenerate principal series

Retain $F = \mathbb{Q}_2$, $M = (\mathbb{Z}_2^4, I_4)$, $\psi_K(a) = \psi(2a)$, and the splitting s_4 fixed above. Let $P_{\mathrm{S}} \subset \mathrm{Sp}_8(F)$ be the upper Siegel parabolic in the original polarization, with

$$m(a) = \mathrm{diag}(a, a^{-T}), \qquad u(b) = \begin{pmatrix} I_4 & b \\ 0 & I_4 \end{pmatrix}, \qquad a \in \mathrm{GL}_4(F), \quad b = b^T.$$

Define normalized induction $I(s)$ by right translation on smooth functions satisfying

$$\varphi(m(a)u(b)g) = \delta_{P_{\mathrm{S}}}(m(a))^{1/2} |\det a|_2^s \varphi(g). \tag{12.7}$$

Conjugation on the unipotent radical sends b to aba^T. Its determinant on $\mathrm{Sym}_4(F)$ is $(\det a)^5$, so

$$\delta_{P_{\mathrm{S}}}(m(a)) = |\det a|_2^5. \tag{12.8}$$

Proposition 12.4 (The evaluation-at-zero embedding)**.** On the realization $\Theta_0 = \Omega^{T(H_0)}$ of Theorem 12.3, the map

$$\Phi : \Theta_0 \longrightarrow I(-1/2), \qquad \Phi(f)(g) = \big(\omega(s_4(g))f\big)(0) \tag{12.9}$$

is a nonzero injective $\mathrm{Sp}_8(F)$-equivariant map. More precisely,

$$\Phi(f)(m(a)u(b)g) = |\det a|_2^2 \Phi(f)(g). \tag{12.10}$$

Thus Θ_0 is realized as an irreducible subrepresentation of $I(-1/2)$, with no additional determinant character.

Proof. For a lift of $m(a)$ in the source metaplectic group, the Levi operator is the change of variables with modulus factor $|\det a|_2^{1/2}$, with scalar $\varepsilon\gamma(\det a, \psi)^{-1}$, $\varepsilon \in \mu_2$. Here γ is the normalized Weil index, whose character is $\psi_K(\frac{1}{2}\,\cdot) = \psi$. The identity $\gamma(c, \psi)^4 = 1$ holds for every $c \in F^\times$ [11, I.4, Lemma 4.1]. Consequently the scalar, including the choice of lift sign, disappears in its fourth tensor power. By the specified tensor splitting of Proposition 11.1, the full Schwartz–Bruhat-space formulas are

$$\big(\omega(s_4(m(a)))f\big)(Z) = |\det a|_2^2 f(Za), \tag{12.11}$$

$$\big(\omega(s_4(u(b)))f\big)(Z) = \psi\big(\mathrm{tr}(Z^T Z b)\big) f(Z). \tag{12.12}$$

These are also the formulas of [11, II.4, Proposition 4.3, pp. 37–38] with ψ_K: the signed-discriminant character of the four-dimensional quadratic space is

$$\chi_{M_F}(c) = (c, (-1)^6 \det(I_4))_F = 1.$$

In particular, the unipotent multiplier is one at zero. Applying the two formulas to $\omega(s_4(g))f$ proves (12.10). Since

$$|\det a|_2^2 = \delta_{P_{\mathrm{S}}}(m(a))^{1/2} |\det a|_2^{-1/2},$$

the image satisfies the transformation law for $I(-1/2)$. Smoothness follows because an open subgroup fixing f also fixes $\Phi(f)$ under right translation. For $g_0 \in \mathrm{Sp}_8(F)$,

$$\Phi\big(\omega(s_4(g_0))f\big)(g) = \Phi(f)(gg_0),$$

which is equivariance for the specified right-translation action. Finally the vector of (12.4) satisfies

$$\Phi(f_+)(1) = f_+(0) = \frac{1}{16} \neq 0.$$

Thus Φ is nonzero. Its kernel is an invariant subspace of the irreducible Θ_0 from Theorem 12.3, and must therefore be zero. □

This is Kudla's evaluation-at-zero construction [11, III.5, p. 50] in the stated normalization. Section 13 identifies its Langlands quotient; Section 14.3 determines the other submodules of $I(-1/2)$.

12.6 A nonintegral Levi element and the two compact types

In the original polarization take $g_2 = \mathrm{diag}(2I_4, \frac{1}{2}I_4) \in \mathrm{Sp}_8(F)$. The source Levi determinant is the square 16, and its lift sign disappears in the fourth tensor power. Thus its exact action is

$$\omega(s_4(g_2))f(Z) = 2^{-8}f(2Z).$$

For f_+ in (12.4), this gives

$$v_+ = 2^{-12}\mathbf{1}_{(D^*)^4}, \qquad (I-P)E_{\mathcal{H}}v_+ = 2^{-12}(\mathbf{1}_{M^4} - \mathbf{1}_{D^4}) \neq 0.$$

For any $v \in \mathcal{M}(\sigma_0)$, the projection onto the embedded negative type is

$$\mathrm{pr}_{\mathcal{A}(F^-)}v = 3\mathcal{A}(I-P)E_{\mathcal{H}}v.$$

Indeed $\mathcal{A}_-^*\mathcal{A}_- = (1/3)I$ and $\mathcal{A}_-^* v = (I-P)E_{\mathcal{H}}v$; the latter lies in F^- because the compression commutes with K_0. Injectivity now shows that the projection of v_+ is nonzero. This explicitly connects the two finite types inside the same local irreducible representation.

13 Langlands identification and Jacquet modules of the local lift

Throughout this section $F = \mathbb{Q}_2$, and parabolic induction and Jacquet functors are normalized. We identify the representation in Proposition 12.4 using [14, 15], then compute its Jacquet modules. The argument originated in [20].

13.1 The ordinary theta tower and its relation to the covers

Let $V = M_F$, $B(x,y) = x^T y$, and $q_V(x) = B(x,x)/2$. The norm $N_V(x) = x^T x$ has the same isometry group $H_0 = O(V)(F)$. For $n \geq 0$, put

$$W_n = F^{2n}, \qquad \Omega_n = \mathcal{S}(V^n), \qquad \mathcal{T}_n = (\Omega_n)_{H_0} = \Theta_{\psi_K, W_n}(\mathbf{1}_{H_0}), \qquad \psi_K(t) = \psi(2t).$$

Here the subscript denotes ordinary orthogonal coinvariants, before any assertion about irreducibility. We use Kudla's even-dimensional ordinary action of $H_0 \times \mathrm{Sp}(W_n)$, with $T_h f(Z) = f(h^{-1}Z)$ on the orthogonal side [11, II.4]. To avoid confusing symplectic rank with the orthogonal label in Section 12, we use the distinct notation $\mathcal{T}_n$:

$$\mathcal{T}_4 = \Theta_0, \qquad \mathcal{T}_0 = \mathbb{C}. \tag{13.1}$$

Here $\mathcal{T}_0$ denotes the rank-zero lift.

Compact averaging identifies E^{H_0} with E_{H_0}, as in the proof of Theorem 12.3. A compact linear group stabilizes a lattice: intersect the finite orbit of any initial lattice, whose stabilizer is open. The characteristic function of its n-th power is a nonzero element of $\Omega_n^{H_0}$. Consequently $\mathcal{T}_n \neq 0$ for every n. Since the trivial representation of the full anisotropic orthogonal group is supercuspidal, [13, Theorem 1.1(i)], equivalently [14, Proposition 2.1(i)], implies that each positive-rank $\mathcal{T}_n$ is irreducible and equals its small theta lift. The cited assertion includes residue characteristic two.

This is an action of the isometry groups, not in general a section of the ambient metaplectic double cover. At $n = 4$ it agrees with s_O, s_4 after removing σ_0. At odd n, a determinant-minus-one reflection has lift scalars $\pm i$, so its normalized operator T_h lies only in the extension by all scalar intertwiners; see Proposition 11.2.

13.2 The tempered inducing representation of Sp_4

Write $\mathbb{D} = (-1,-1)_F$ for the quaternion division algebra; its reduced norm in the standard basis is N_V. The symbol $\mathbb{D}$ is distinct from both the lattice D of Section 12 and the integer $n(2n+1)$. Let Q_{Kl} be the Klingen parabolic of $\mathrm{GSp}_4(F)$, whose Levi is $\mathrm{GL}_1(F) \times \mathrm{GL}_2(F)$, and define

$$\Pi_{\mathrm{ng}} = \pi_{\mathrm{ng}}(\mathrm{St}_{\mathrm{GL}_2}), \qquad \tau_{\mathrm{ng}} = \mathrm{Res}^{\mathrm{GSp}_4(F)}_{\mathrm{Sp}_4(F)} \Pi_{\mathrm{ng}}.$$

Here $\mathrm{St}_{\mathrm{GL}_2}$ is the untwisted Steinberg representation, and Π_{ng} is the unique nongeneric tempered summand of

$$\mathrm{Ind}^{\mathrm{GSp}_4(F)}_{Q_{\mathrm{Kl}}}(\mathbf{1} \boxtimes \mathrm{St}_{\mathrm{GL}_2}).$$

The direct-sum description is [14, Lemma 5.1(b)]. The next proposition proves that the restriction itself is irreducible.

Proposition 13.1 (The tempered inducing representation)**.** There is an isomorphism of ordinary $\mathrm{Sp}_4(F)$-representations

$$\tau_{\mathrm{ng}} \simeq \mathcal{T}_2.$$

In particular, the displayed restriction is irreducible and tempered.

Proof. First the ordinary big theta lift of $\det_{H_0}$ is zero in symplectic ranks $n < 4$. Indeed, a tuple in V^n spans a proper nondegenerate subspace U, because V is anisotropic. Choose $0 \neq u \in U^{\perp}$. The reflection

$$r_u(x) = x - 2\frac{B(x,u)}{B(u,u)}u$$

fixes the tuple and has determinant -1. A function transforming by $\det_{H_0}$ therefore vanishes at that tuple. It vanishes everywhere, and compact averaging identifies this zero isotypic space with the corresponding big theta multiplicity space.

The two extensions of the trivial $\mathrm{GSO}(\mathbb{D})$-representation to $\mathrm{GO}(\mathbb{D})$ are $\mathbf{1}$ and

$$\epsilon_{\mathrm{GO}}(h) = \det(h)\lambda_V(h)^{-2}.$$

They restrict to $\mathbf{1}_{H_0}$ and $\det_{H_0}$, respectively. The exponent -2 uses $\dim V = 4$. By [14, Lemma 2.2(iii)], the big similitude lifts restrict exactly to these ordinary big lifts. Thus the ϵ_{GO} lift to GSp_4 is zero, whereas the trivial-extension lift restricts to $\mathcal{T}_2$. The quotient for the trivial GSO-type splits into the two extension quotients under $\mathrm{GO}\,/\,\mathrm{GSO} \simeq C_2$; only the trivial extension contributes.

Under

$$\mathrm{GSO}(\mathbb{D}) \simeq (\mathbb{D}^\times \times \mathbb{D}^\times)/\{(z, z^{-1}) : z \in F^\times\},$$

the input is $\mathbf{1} \boxtimes \mathbf{1}$. The Jacquet–Langlands normalization sends $\mathbf{1}_{\mathbb{D}^\times}$ to $\mathrm{St}_{\mathrm{GL}_2}$. Therefore [14, Theorem 8.1(ii)] identifies its similitude lift with Π_{ng}, and the exact restriction comparison gives $\tau_{\mathrm{ng}} \simeq \mathcal{T}_2$. Irreducibility follows from the compact-input argument above, not from a general assertion about restrictions from GSp_4 to Sp_4.

Temperedness also follows independently from [15, Proposition 5.5(1)]: its input is the tempered trivial representation of the compact orthogonal group, and its full dimensions are $n_{\mathrm{AG}} = m_{\mathrm{AG}} = 4$, with $\epsilon_0 = -1$. Thus $m_{\mathrm{AG}} \leq n_{\mathrm{AG}} + 1 + \epsilon_0$, as required. □

Scaling the norm in a source convention changes the additive character in the ordinary Weil model. This does not change the identification just made: reduced norms are surjective, so the similitude factor on $\mathrm{GSO}(\mathbb{D})$ is surjective, and the similitude Weil representation is independent of the nontrivial additive character [14, Section 2, pp. 4–5]. Lemma 2.2(iii) of that source therefore applies to the particular ψ_K fixed here. This observation concerns the isomorphism class of the ordinary lift; it does not remove the specified finite or ambient Weil phases.

13.3 The standard module for the lift to Sp_8

Write $\nu = |\cdot|_F$, and let $P_{1,1} \subset \mathrm{Sp}_8(F)$ be the upper parabolic stabilizing an isotropic flag of dimensions one and two, with Levi $\mathrm{GL}_1 \times \mathrm{GL}_1 \times \mathrm{Sp}_4$. For this parabolic induction has the same right-translation normalization as (12.7).

Theorem 13.2 (Langlands quotient of the trivial-representation lift)**.** The ordinary representation Θ_0 of Theorem 12.3 is

$$\Theta_0 \simeq L(\nu^2, \nu; \tau_{\mathrm{ng}}), \tag{13.2}$$

where the right side denotes the unique irreducible quotient of the standard module

$$\mathrm{Ind}_{P_{1,1}}^{\mathrm{Sp}_8(F)}(\nu^2 \boxtimes \nu \boxtimes \tau_{\mathrm{ng}}). \tag{13.3}$$

Consequently $\Phi(\Theta_0) \subset I(-1/2)$ is a copy of this specified Langlands quotient.

Proof. Apply [15, Proposition 5.6], with the roles of the named spaces matched explicitly: its W is our four-dimensional orthogonal input, and its V is the symplectic target. In its notation,

$$E = F, \qquad \epsilon = \epsilon_0 = -1, \qquad n_{\mathrm{AG}} = m_{\mathrm{AG}} = 4, \qquad l = n_{\mathrm{AG}} - m_{\mathrm{AG}} + \epsilon_0 = -1.$$

These are full dimensions, not symplectic ranks. The discriminant character χ_W is trivial. The initial lift is the nonzero tempered $\mathcal{T}_2 = \tau_{\mathrm{ng}}$ of Proposition 13.1. Adding $r = 2$ symplectic hyperbolic planes reaches Sp_8, and the exponents in that proposition are

$$\frac{2r-1-l}{2} = 2, \qquad \frac{1-l}{2} = 1.$$

It identifies the higher small lift with the unique irreducible quotient of (13.3). The big and small lifts here agree, so this is the actual $\mathcal{T}_4 = \Theta_0$. The positive decreasing exponents and the tempered inducing representation make the inducing data a standard module. Finally apply Proposition 12.4. □

13.4 Normalized Jacquet modules

For $1 \le k \le n$, let $Q_k \subset \mathrm{Sp}(W_n)$ be the upper parabolic stabilizing the first k isotropic coordinates. Its Levi is $\mathrm{GL}_k \times \mathrm{Sp}(W_{n-k})$, its unipotent radical is U_k, and $R_{Q_k}E = \delta_{Q_k}^{-1/2} \otimes E_{U_k}$.

Lemma 13.3 (Compact averaging and Jacquet modules)**.** For a smooth representation E of the commuting groups H_0 and $\mathrm{Sp}(W_n)$,

$$R_{Q_k}(E^{H_0}) \simeq (R_{Q_k}E)^{H_0}.$$

Proof. The decomposition $E = e_H E \oplus \ker e_H$ is Q_k-stable. Taking U_k-coinvariants preserves this direct sum, and the induced projector is still Haar averaging for H_0. Its image is therefore the full H_0-fixed space of the quotient. The normalizing modulus is trivial on H_0. □

Proposition 13.4 (Normalized Jacquet modules)**.** For $1 \le k \le n$, there are isomorphisms

$$R_{Q_k}(\Omega_n) \simeq |\det|_F^{(k+3-2n)/2} \boxtimes \Omega_{n-k}, \tag{13.4}$$

$$R_{Q_k}(\mathcal{T}_n) \simeq |\det|_F^{(k+3-2n)/2} \boxtimes \mathcal{T}_{n-k}. \tag{13.5}$$

The first is an isomorphism for $\mathrm{GL}_k(F) \times \mathrm{Sp}(W_{n-k}) \times H_0$, and the second for the first two factors. These are full modules, not only semisimplifications or selected quotients of a filtration.

Proof. Kudla's filtration, in the normalization of [14, Appendix A, Theorem A.2 and Remark A.6], has layers indexed by $0 \le t \le \min(k, r_V)$. The Witt index is $r_V = 0$, so only the zero layer remains. The following direct calculation also verifies its exponent and the assertion about the whole module.

Write the coordinates as $(X, Y) \in V^k \times V^{n-k}$ and consider restriction to $X = 0$. The subgroup of U_k corresponding to symmetric upper matrices supported on the first k coordinates acts by

$$f(X, Y) \longmapsto \psi\big(\mathrm{tr}(X^T X b)\big) f(X, Y), \qquad b = b^T \in \mathrm{Mat}_k(F).$$

Its common trivial-character locus is $X^T X = 0$. Anisotropy forces each column of X to be zero, so that locus is exactly $X = 0$. If a Schwartz–Bruhat function restricts to zero there, its compact support is disjoint from that locus. At each point of the support one can choose a parameter b with nontrivial phase, and then a compact open neighborhood on which that phase is a fixed scalar different from one. A finite disjoint refinement of such neighborhoods writes the function as a sum of differences $u \cdot f - f$. Thus restriction identifies the coinvariants for this subgroup with $\mathcal{S}(V^{n-k})$.

The remaining generators of U_k act trivially on this restriction: their mixed changes of variables and phases vanish when the first k coordinates are zero, by the same root formulas as (2.2). Restriction is surjective and hence also identifies the full unnormalized U_k-coinvariants with Ω_{n-k}. The GL_k-action before normalization is $|\det a|_F^2$, since $\dim V = 4$ and $\chi_V = 1$. The remaining symplectic and orthogonal actions are their ordinary Weil actions.

The roots in U_k give

$$\delta_{Q_k}(a) = |\det a|_F^{2n-k+1}.$$

For a diagonal entry of a, the exponent is $2(n-k) + (k-1) + 2 = 2n - k + 1$: the terms come respectively from mixed roots, roots joining two of the first k coordinates, and the long root. Therefore the normalized exponent is

$$2 - \frac{2n - k + 1}{2} = \frac{k + 3 - 2n}{2}.$$

This proves (13.4), in agreement with the single layer of the cited filtration. Apply Lemma 13.3 and $\mathcal{T}_j \simeq \Omega_j^{H_0}$ to obtain (13.5), including $j = 0$. □

The useful specializations are

(n, k)	$R_{Q_k}(\mathcal{T}_n)$	Levi
$(4, 1)$	$\nu^{-2} \boxtimes \mathcal{T}_3$	$\mathrm{GL}_1 \times \mathrm{Sp}_6$
$(3, 1)$	$\nu^{-1} \boxtimes \tau_{\mathrm{ng}}$	$\mathrm{GL}_1 \times \mathrm{Sp}_4$
$(4, 2)$	$\lvert\det\rvert^{-3/2} \boxtimes \tau_{\mathrm{ng}}$	$\mathrm{GL}_2 \times \mathrm{Sp}_4$
$(4, 4)$	$\lvert\det\rvert^{-1/2}$	GL_4

In the third row, the normalized upper-Borel Jacquet module of $|\det|^{-3/2}$ on GL_2 has exponents $-2, -1$, in agreement with the first two rows and Jacquet transitivity. These upper-parabolic embedding exponents must be distinguished from the positive Langlands-quotient data in (13.3); adjunction for a quotient uses the opposite parabolic. The induction parameter in the specified Siegel embedding remains $-1/2$.

Corollary 13.5 (The specified embedding space)**.** With $P_{\mathrm{S}} = Q_4 \subset \mathrm{Sp}_8(F)$,

$$R_{P_{\mathrm{S}}}(\Theta_0) \simeq |\det|_F^{-1/2}, \qquad \dim_{\mathbb{C}} \mathrm{Hom}_{\mathrm{Sp}_8(F)}(\Theta_0, I(-1/2)) = 1. \tag{13.6}$$

Thus the evaluation map realizes the unique embedded copy of $L(\nu^2, \nu; \tau_{\mathrm{ng}})$ in this induction, with the embedding unique up to a nonzero scalar.

Proof. The Jacquet assertion is the last row of the table. For the right-translation convention of (12.7), normalized Frobenius reciprocity uses this same upper parabolic and gives

$$\mathrm{Hom}_{\mathrm{Sp}_8(F)}(\Theta_0, I(-1/2)) \simeq \mathrm{Hom}_{\mathrm{GL}_4(F)}(|\det|^{-1/2}, |\det|^{-1/2}) \simeq \mathbb{C}.$$

The nonzero map Φ already exists by Proposition 12.4. Combine this with Theorem 13.2. □

This uniqueness concerns the specified isomorphism class. The full socle of $I(-1/2)$ is determined in Theorem 14.2. The compact types constructed above remain subrepresentations of the same local lift; their full restriction multiplicities are not determined here.

14 Theta lifts over dyadic fields and the Siegel principal series

Let $F/\mathbb{Q}_2$ be finite and $\nu_F = |\cdot|_F$. We first distinguish scalar extension of I_4 from the choice of an anisotropic norm over F, then identify both theta lifts in the same Siegel degenerate principal series. All parabolic inductions and Jacquet functors are normalized.

14.1 Scalar extension of the form I_4

Put $d = [F : \mathbb{Q}_2]$ and $N_V(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2$. Write $\mathbb{D}_0 = (-1, -1)_{\mathbb{Q}_2}$.

Proposition 14.1 (The total-degree criterion)**.** The scalar extension of I_4 is anisotropic over F if and only if d is odd. For even d it is hyperbolic of dimension four and has Witt index two.

Proof. The quaternion algebra of the norm is $\mathbb{D}_0 \otimes_{\mathbb{Q}_2} F$. Since $\mathbb{D}_0$ is division, its local Brauer invariant is $1/2$. Restriction of local invariants gives

$$\mathrm{inv}_F(\mathbb{D}_0 \otimes_{\mathbb{Q}_2} F) = d\,\mathrm{inv}_{\mathbb{Q}_2}(\mathbb{D}_0) = \frac{d}{2} \pmod{\mathbb{Z}}. \tag{14.1}$$

This formula holds also for non-Galois extensions [16, III.2, equation (29), p. 105]. For odd d the algebra remains division, whose norm is anisotropic. For even d it is $\mathrm{Mat}_2(F)$, and its norm is the determinant, a hyperbolic quadratic form in four variables. Scaling between N_V and $q_V = N_V/2$ does not change either conclusion. □

14.2 An anisotropic quaternion norm over each field

Let $\tau_{\mathrm{ng},F}$ be defined over F by the construction of τ_{ng} in Section 13.2. Choose the quaternion division algebra $\mathbb{D}_F$ over F, with its norm space and the ordinary even-dimensional Weil action. The norm has trivial signed discriminant and compact isometry group. Repeating the compact averaging, reflection, similitude comparison, and higher-lift arguments proves, for every finite $F/\mathbb{Q}_2$,

$$\Theta_{\psi_K, \mathrm{Sp}_8(F)}(\mathbf{1}_{O(\mathbb{D}_F, N_{\mathbb{D}_F})}) \simeq L(\nu_F^2, \nu_F; \tau_{\mathrm{ng},F}). \tag{14.2}$$

The full Jacquet formula and the one-dimensional embedding space in $I_F(-1/2)$ hold for this anisotropic input as well. Kudla's ordinary even-dimensional splitting is used throughout. If this norm is not isometric to I_4, it must not be realized as four identical I_1 tensor factors without changing the form.

14.3 The three constituents of the Siegel principal series

Put $I_F^- = I_F(-1/2)$, $I_F^+ = I_F(1/2)$, with the upper Siegel parabolic and right-translation convention of (12.7). Let $V_{4,F}^{\mathrm{sp}}$ be the split four-dimensional quadratic space and $V_{4,F}^{\mathrm{an}}$ the norm space of

the quaternion division algebra over F. Both have trivial signed-discriminant character. Define the big theta lifts of their trivial representations by

$$R_F(V) = \mathcal{S}(V^4)_{O(V)}, \qquad \Pi_{\mathrm{sp},F} = R_F(V_{4,F}^{\mathrm{sp}}), \qquad \Pi_{\mathrm{an},F} = R_F(V_{4,F}^{\mathrm{an}}).$$

For the split space the orthogonal group is noncompact, so the coinvariant realization is essential.

Let $\pi_{0,F} = \mathrm{Ind}_B^{\mathrm{Sp}_2(F)}(\mathbf{1})$ be normalized induction. This unitary principal series is irreducible and tempered; for this inducing line the reducibility points are $\nu_F^{\pm 1}$, not $\mathbf{1}$. Let $\mathrm{St}_{2,F}$ denote the unitary Steinberg representation of $\mathrm{GL}_2(F)$. In the positive-exponent, unique-quotient convention, set

$$\begin{aligned} A_F &= L(\nu_F^2, \nu_F, \nu_F; \pi_{0,F}), \\ B_F &= L(\nu_F^2, \nu_F; \tau_{\mathrm{ng},F}), \\ X_F &= L(\nu_F^{3/2}\mathrm{St}_{2,F}; \tau_{\mathrm{ng},F}). \end{aligned} \tag{14.3}$$

The corresponding Levis are respectively $\mathrm{GL}_1^3 \times \mathrm{Sp}_2$, $\mathrm{GL}_1^2 \times \mathrm{Sp}_4$, and $\mathrm{GL}_2 \times \mathrm{Sp}_4$. The notation A_F, B_F here denotes representations, not the finite rings or forms used in earlier sections.

Theorem 14.2 (The three constituents and the complete submodule lattice)**.** The two big theta lifts above are irreducible and satisfy

$$\Pi_{\mathrm{sp},F} \simeq A_F, \qquad \Pi_{\mathrm{an},F} \simeq B_F.$$

Evaluation at zero embeds each of them into I_F^-. Their images are disjoint, and there is an exact sequence

$$0 \longrightarrow A_F \oplus B_F \longrightarrow I_F^- \longrightarrow X_F \longrightarrow 0. \tag{14.4}$$

All three composition factors are pairwise inequivalent and occur once. The submodules of I_F^- are exactly

$$0, \quad A_F, \quad B_F, \quad A_F \oplus B_F, \quad I_F^-.$$

In particular, its socle and radical are both $A_F \oplus B_F$, and its cosocle is X_F. For $F = \mathbb{Q}_2$, the B_F submodule is the image of the specified Φ in Proposition 12.4.

Proof. The Rallis embedding, as stated for the *full* orthogonal coinvariant quotient in [17, Section 4, p. 23], is

$$R_F(V) \hookrightarrow I_F\big(\dim V/2 - 5/2, \chi_V\big), \qquad f \longmapsto [g \mapsto (\omega(g)f)(0)].$$

That source assumes a nonarchimedean field of characteristic zero and does not exclude residue characteristic two. Its parameter $2n$ is the symplectic rank of the doubled group; here $2n = 4$, not 8. In its formulas (4.5), (4.7), and (4.8), take $m_0 = 4$, $m = 6$, $s_0 = 1/2$, and $\chi = 1$. The two existing $R_4(V_4^{\pm})$ are irreducible and disjoint, their sum is a maximal proper submodule of I_F^-, and the quotient is irreducible. The parameter is $4/2 - 5/2 = -1/2$. These statements concern the usual even-dimensional Weil action. Changing the nontrivial additive character does not change the isomorphism classes here: scalar multiples of either the split norm or the division-quaternion norm are isometric, since their similitude multipliers are all of $F^\times$. No finite Weyl operator is rephased in making this assertion.

For independent constituent labels and the cosocle assertion, specialize [18, Proposition 3.10(3)(a); Theorems 6.2(3c), 7.2(3c)] to

$$\rho = \mathbf{1}_{\mathrm{GL}_1}, \quad \sigma = \mathbf{1}_{\mathrm{Sp}_0}, \quad k = 4, \quad \ell = 0, \quad \alpha = -1/2 = (-4+1)/2 + 1.$$

Thus $j = 1$. His $\zeta(\mathbf{1}, 4)$ is $\mathbf{1}_{\mathrm{GL}_4}$, $\delta(\mathbf{1}, 2) = \mathrm{St}_{2,F}$, and brackets such as $[\nu_F^{-2}, \nu_F^{-1}]$ in the Langlands arguments list two GL_1 factors: they are not a single Steinberg factor. He uses a negative-exponent *subrepresentation* convention. The three lists there become respectively

$$(\nu_F^{-2}, \nu_F^{-1}, \nu_F^{-1}; \pi_{0,F}), \quad (\nu_F^{-2}, \nu_F^{-1}; T), \quad (\nu_F^{-3/2}\mathrm{St}_{2,F}; T).$$

Here T is the unique common irreducible constituent of the Sp_4 Siegel induction $\nu_F^{-1/2}\mathbf{1}_{\mathrm{GL}_2} \rtimes \mathbf{1}$ and the Klingen induction $\mathbf{1} \rtimes \mathrm{St}_{\mathrm{Sp}_2}$. It equals $\tau_{\mathrm{ng},F}$. Indeed the division-norm lift to Sp_4 is that nongeneric Klingen constituent by the comparison already proved in Section 13; evaluation at zero places it in the Siegel induction with parameter $+1/2$, which has the same composition factors as the induction with parameter $-1/2$. Uniqueness of the common constituent identifies T. Moreover T is self-dual: taking contragredients preserves the two sets of common constituents, and the common constituent is unique. The same is true of $\pi_{0,F}$. Lemma 2.9(3) of that source then makes each negative-list module self-dual. Taking its contragredient converts the displayed negative lists into exactly the positive-quotient labels in (14.3); compare [18, Section 2, pp. 5–6; Lemma 2.9].

Jantzen's composition-series theorem gives $A_F \oplus B_F$ as a subrepresentation and X_F as the unique irreducible quotient, with all factors occurring once. The already identified anisotropic lift is B_F, so the other Rallis submodule is A_F. This also proves that each submodule in the statement is its indicated, unique copy. If a proper submodule were not contained in $A_F \oplus B_F$, its image in X_F would be all of X_F. The remaining nonzero quotient would then have an irreducible quotient of type A_F or B_F, contradicting the unique cosocle. All proper submodules are therefore contained in the semisimple, multiplicity-free sum, giving the asserted full lattice. □

14.4 Intertwining operators and nonsplit extensions

Let $M_+^* : I_F^+ \to I_F^-$ and $M_-^* : I_F^- \to I_F^+$ be the normalized operators at $s = 1/2, -1/2$ in [17, Section 4]. Their kernel–image identities are

$$\operatorname{im} M_+^* = \ker M_-^* = A_F \oplus B_F, \qquad \operatorname{im} M_-^* = \ker M_+^* \simeq X_F. \tag{14.5}$$

Thus I_F^+ has the dual, reversed submodule lattice. In particular, X_F is its irreducible socle, not a maximal proper submodule. This distinction follows both from Jantzen's theorem and from formulas (4.10)–(4.12) of [17]; it is needed when reading the sentence following (4.8) in the author-hosted version of that source.

Quotienting (14.4) by either summand gives

$$0 \to A_F \to I_F^-/B_F \to X_F \to 0, \qquad 0 \to B_F \to I_F^-/A_F \to X_F \to 0.$$

Both are nonsplit, since a splitting would give an additional irreducible quotient of I_F^-. They exhibit nonzero classes in $\mathrm{Ext}^1(X_F, A_F)$ and $\mathrm{Ext}^1(X_F, B_F)$; they do not compute the dimensions of those Ext spaces in the ambient smooth category.

14.5 Consequences for scalar extension of I_4

For odd d, put $H_F = O(F^4, I_4)$ and $\mathcal{T}_{n,F} = \Theta_{\psi_K, W_n}(\mathbf{1}_{H_F})$, with $\nu_F = |\cdot|_F$. The compact orthogonal-group argument gives

$$\mathcal{T}_{4,F} \simeq L(\nu_F^2, \nu_F; \tau_{\mathrm{ng},F}), \tag{14.6}$$

$$R_{Q_k}(\mathcal{T}_{n,F}) \simeq |\det|_F^{(k+3-2n)/2} \boxtimes \mathcal{T}_{n-k,F}, \tag{14.7}$$

$$\dim \mathrm{Hom}_{\mathrm{Sp}_8(F)}(\mathcal{T}_{4,F}, I_F(-1/2)) = 1. \tag{14.8}$$

A characteristic function of an H_F-stable lattice is nonzero at zero, so evaluation gives the injective embedding. These statements hold for any nontrivial ψ_K; the conductor of the finite lattice model is a separate choice.

For even d, the fixed form is split. Theorem 14.2 identifies its big theta lift with A_F: it is irreducible, equals its small lift, and the evaluation-at-zero map is injective. This conclusion uses the Rallis and Jantzen structure theorems, not compact averaging. The orthogonal group

is noncompact and the Weil Jacquet filtration has additional layers, so neither the invariant realization nor the single-layer proof of (13.5) applies.

The parity of $d = ef$ here is distinct from the parity of the ramification index e in Theorem 3.3. These local representation-theoretic results do not determine the integral model: the lattice, conductor, orthogonal reduction and finite multiplicities require separate calculation. In particular, the dimensions $51, 3825$ and the index-three lattice stabilizer over $\mathbb{Q}_2$ do not carry over unchanged.

15 Orthogonal averaging on multiplicity spaces

Return now to $F = \mathbb{Q}_2$, $V = (F^4, I_4)$, $H = O(V)$, and $K = O(M)$. Retain the intrinsic lattices D, D^* and the elements t, r of Section 12. In this and the next two sections n is the symplectic rank of the target; the quadratic dimension stays four. Write

$$\Omega_n = \mathcal{S}(V^n), \quad \mathcal{H}_{n,4} = \mathcal{S}(M^n/(2M)^n), \quad \Pi_n = P_n^{(4)}, \quad C_n = \Pi_n \mathcal{H}_{n,4} = \mathcal{S}(D^n/(2D^*)^n).$$

Thus Π_n abbreviates the projection $P_n^{(4)}$, while P_n continues to denote the Pauli group. We use the ordinary orthogonal operators T_h; the genuine central characters are specified below for the lift to Sp_8.

15.1 Two different quotients of the compact orthogonal group

The actions on the two intrinsic four-element quotients give homomorphisms

$$\alpha_H : H \longrightarrow \mathrm{GL}(D/2D^*) \simeq S_3, \qquad \beta_H : H \longrightarrow \mathrm{Perm}\{M_0, M_+, M_-\} \simeq S_3.$$

The second is equivalently the action on the three nonzero elements of D^*/D. It must not be confused with the first, which acts on $D/2D^*$.

Proposition 15.1 (The two quotient actions and determinant)**.** The map $(\alpha_H, \beta_H) : H \to S_3 \times S_3$ is surjective. On K, $\alpha_H = p \circ \mathrm{red}_2$, with the quotient $p : S_4 \to S_3$ of Section 10, whereas $\beta_H(K)$ is the order-two stabilizer of M_0. Furthermore,

$$\det h = \mathrm{sgn}(\alpha_H(h)) \, \mathrm{sgn}(\beta_H(h)). \tag{15.1}$$

In particular, $\mathrm{sgn} \circ \alpha_H \neq \det_H$.

Proof. The space $D/2D^*$ is the previous $N^\perp/N$. Its nonzero classes have norm two modulo four, and the induced action of a coordinate permutation is p. These permutations fix all three lattices M_i, giving $S_3 \times 1$ in the joint image. The reflection r is trivial on $D/2D^*$ and exchanges M_+ with M_-. The element t exchanges M_0, M_+ and, in the basis $[1100], [1010]$, exchanges the two basis vectors of $D/2D^*$. Removing its first component by a coordinate permutation supplies the other transposition of the second S_3. This proves surjectivity. The stabilizer of M_0 is precisely K.

For the determinant formula, write an integral isometry as $h = P_0(I + 2A_0)$, where P_0 is the permutation matrix supplied by its reduction. Orthogonality gives $A_0 \equiv A_0^T \pmod 2$. Since $\det h = \pm 1$, reduction of its determinant modulo four gives

$$\det h = \mathrm{sgn}(P_0)(-1)^{\mathrm{tr}\, A_0}.$$

Writing $v = \mathbf{1}/2$, one has $hv = v + P_0 A_0 \mathbf{1}$. Thus the nontrivial exchange of M_+, M_- occurs exactly when $\mathbf{1}^T A_0 \mathbf{1}$ is odd. Symmetry modulo two makes this parity $\mathrm{tr}\, A_0$. Also $\mathrm{sgn}(p(P_0)) = \mathrm{sgn}(P_0)$. This proves (15.1) on K. It holds on t because $\det t = 1$ and both quotient actions are transpositions. As $H = \langle K, t \rangle$, it holds on H. Finally r has determinant minus one but trivial first quotient. □

For $\sigma \in \{\mathbf{1}, \mathrm{std}, \mathrm{sgn}\} = \mathrm{Irr}\, S_3$, choose a unitary model E_σ and set

$$a_\sigma = \sigma \circ \alpha_H, \qquad \rho_\sigma = a_\sigma|_K = p^*\sigma, \qquad (\lambda_{\mathbf{1}}, \lambda_{\mathrm{std}}, \lambda_{\mathrm{sgn}}) = ([4], [22], [1111]).$$

These are irreducible H-representations extending the indicated K-representations.

15.2 The averaging map and its Gram operator

Let

$$F_{\sigma,n} = \mathrm{Hom}_K(E_\sigma, \mathcal{H}_{n,4}), \qquad F^{\pm}_{\sigma,n} = \mathrm{Hom}_K(E_\sigma, \Pi_n^{\pm}\mathcal{H}_{n,4}), \quad \Pi_n^+ = \Pi_n, \quad \Pi_n^- = I - \Pi_n.$$

Give Hom spaces the Hilbert–Schmidt inner product induced by the fixed unitary model and Haar measure on V^n. Projectors on the target act by postcomposition. Define

$$\mathcal{A}_{\sigma,n}(T) = \frac{1}{3} \sum_{h \in \{1,t,rt\}} T_h\, T\, a_\sigma(h)^{-1}. \tag{15.2}$$

Let E_n be the orthogonal projection of $L^2(V^n)$ onto $\mathcal{H}_{n,4}$, also acting by postcomposition.

Theorem 15.2 (Compression of the averaging map)**.** The integral symplectic action commutes with $\mathcal{A}_{\sigma,n}$, whose values lie in $\mathrm{Hom}_H(E_\sigma, \Omega_n)$. On its finite-dimensional source $F_{\sigma,n}$, this map satisfies

$$E_n\mathcal{A}_{\sigma,n} = \mathcal{A}^*_{\sigma,n}\mathcal{A}_{\sigma,n} = \Pi_n + \frac{1}{3}(I - \Pi_n). \tag{15.3}$$

It is injective on both nonzero summands. The $+$-summand is unchanged; the $-$-summand has squared-norm factor $1/3$, has image orthogonal to the image of the $+$-summand, and intersects $\mathrm{Hom}(E_\sigma, \mathcal{H}_{n,4})$ trivially. At $n = 4$, the two dimensions are respectively

$$(51, 3825), \qquad (85, 5355), \qquad (35, 1785)$$

for $\sigma = \mathbf{1}, \mathrm{std}, \mathrm{sgn}$.

Proof. On $\mathrm{Hom}(E_\sigma, \Omega_n)$, the action is $h \cdot T = T_h T a_\sigma(h)^{-1}$. Because T is K-fixed for this action, compact averaging reduces to the three left cosets and gives exactly (15.2). It is the restriction of an orthogonal projection and commutes with the symplectic group whenever the latter preserves the chosen finite source.

The code C_n is H-stable, and its action is the permutation action of α_H on $(D/2D^*)^n$. Since $\alpha_H(K) = \alpha_H(H)$, a K-map into C_n is already an H-map. Thus $\mathcal{A}_{\sigma,n}\Pi_n = \Pi_n$. The one-column calculation in Theorem 12.2 tensors to any n and gives

$$E_n T_h|_{\mathcal{H}_{n,4}} = T_h \Pi_n, \qquad h = t, rt.$$

Postcomposing (15.2) with E_n therefore gives $(T + 2\Pi_n T)/3$. The adjoint identity follows from orthogonal averaging on the Hom space. Its two positive eigenvalues prove all the norm, injectivity, and orthogonality assertions. If a negative averaged map lies in the finite space, compression makes it $T/3$; averaging again forces it to be one third of itself, hence zero. The dimensions are the previously proved multiplicities, not new dimension assumptions. □

Corollary 15.3 (Three local theta lifts and their central characters)**.** At $n = 4$, both displayed compact types for each σ generate one nonzero irreducible local representation

$$\mathcal{T}_4(a_\sigma) = \mathrm{Hom}_H(a_\sigma, \Omega_4) \simeq \Theta_{\psi_K}(a_\sigma) = \theta_{\psi_K}(a_\sigma).$$

The three local representations are pairwise inequivalent. For the full orthogonal inverse image, the input is explicitly $\widetilde{h} \mapsto \sigma_0(\widetilde{h})a_\sigma(p(\widetilde{h}))$; its integral restriction is $\chi_2 \boxtimes \rho_{\lambda_\sigma}$. The corresponding complete symplectic action is genuine.

Proof. Every irreducible smooth representation of the compact anisotropic H is supercuspidal. Compact semisimplicity identifies the Hom realization with $(\Omega_4 \otimes a_\sigma^\vee)_H$. Nonvanishing follows from the theorem; the supercuspidal-input assertion of [13] makes big and small theta equal and irreducible. Howe injectivity distinguishes the three inputs. The last statements follow by multiplying back the already fixed scalar character, not by a new choice of phases. □

16 The subrepresentation generated by orthogonal translates

Let $\mathscr{V}_n$ be the H-subrepresentation of Ω_n generated by the finite lattice model:

$$\mathscr{V}_n = \sum_{h\in H} T_h \mathcal{H}_{n,4} \subset \Omega_n, \qquad \mathcal{H}_{n,4}^- = (I - \Pi_n)\mathcal{H}_{n,4}.$$

Proposition 16.1 (Decomposition of the generated subrepresentation)**.** There is an orthogonal decomposition and an $H \times K_n$-module isomorphism

$$\mathscr{V}_n = C_n \ \perp\ \mathcal{H}_{n,4}^- \ \perp\ T_t\mathcal{H}_{n,4}^- \ \perp\ T_{rt}\mathcal{H}_{n,4}^- \simeq C_n \oplus \operatorname{Ind}_K^H(\mathcal{H}_{n,4}^-). \tag{16.1}$$

Here induction is finite-index induction, with the sum inner product on its three coset fibers. In particular,

$$\dim \mathscr{V}_n = 4^n + 3(16^n - 4^n), \qquad \dim \mathscr{V}_4 = 196096.$$

Proof. The three cosets exhaust H/K; the code is H-stable. The compression formula kills $\mathcal{H}_{n,4}^-$ whenever the translating element lies outside K. Indeed K has just two double cosets in H: K and KtK, since $\beta_H(K)$ exchanges the other two lattices. For k_1tk_2, insert the two integral operators on either side of $E_nT_t = T_t\Pi_n$. This proves the pairwise orthogonality of the three negative translates. Their direct sum is the finite-index induced representation. The decomposition is $H \times K_n$-equivariant. □

Define three further orthogonal representations

$$b_\sigma = (\sigma \boxtimes \mathrm{std}) \circ (\alpha_H, \beta_H), \qquad \sigma \in \{\mathbf{1}, \mathrm{std}, \mathrm{sgn}\},$$

and two more by

$$\varrho_{31} = \operatorname{Ind}_K^H \rho_{[31]}, \qquad \varrho_{211} = \operatorname{Ind}_K^H \rho_{[211]}.$$

All integral S_4-representations here are inflated along reduction.

Theorem 16.2 (Induction from the lattice stabilizer)**.** The eight representations

$$\mathfrak{X} = \{a_{\mathbf{1}}, a_{\mathrm{std}}, a_{\mathrm{sgn}}, b_{\mathbf{1}}, b_{\mathrm{std}}, b_{\mathrm{sgn}}, \varrho_{31}, \varrho_{211}\}$$

are irreducible and pairwise inequivalent, of dimensions $1, 2, 1, 2, 4, 2, 9, 9$. Moreover,

$$\operatorname{Ind}_K^H \rho_{\lambda_\sigma} \simeq a_\sigma \oplus b_\sigma, \qquad \operatorname{Ind}_K^H \rho_{[31]} = \varrho_{31}, \quad \operatorname{Ind}_K^H \rho_{[211]} = \varrho_{211}. \tag{16.2}$$

Proof. Surjectivity onto $S_3 \times S_3$ proves irreducibility and inequivalence of the six displayed product-quotient representations. The projection formula gives

$$\operatorname{Ind}_K^H(a_\sigma|_K) = a_\sigma \otimes \mathbb{C}[H/K] = a_\sigma \otimes (\mathbf{1} \oplus \mathrm{std} \circ \beta_H),$$

which proves their induction formula.

For the other two types, use the two double cosets above and put $J = K \cap tKt^{-1}$. The Klein four group of coordinate double transpositions lies in J. Conjugation by the displayed Hadamard matrix t sends each of these permutation matrices to a diagonal sign matrix. Thus its conjugate reduces to the identity modulo two. In contrast, the restriction of either $\rho_{[31]}$ or

$\rho_{[211]}$ to that Klein four group is the sum of its three nontrivial characters and has no fixed vector. Consequently the nonidentity double-coset intertwining space

$$\mathrm{Hom}_J(\rho_\lambda, \rho_\mu^t) = 0, \qquad \lambda, \mu \in \{[31], [211]\}.$$

The finite-index Mackey decomposition of Hom spaces now gives dimension one for the endomorphism algebra of either induced representation and zero between them. Compact semisimplicity proves the assertions. Finally the same Klein four group acts trivially in all six product- quotient representations, so Frobenius reciprocity excludes equivalence between either nine-dimensional representation and any of those six.

□

For convenience write $M^{\pm}_{n,\lambda} = \Theta^{\pm}_{n,4}(\rho_\lambda)$, allowing a zero multiplicity space. Evaluation, the preceding theorem, and (16.1) give the entire joint decomposition

$$\begin{aligned} \mathscr{V}_n \simeq \bigoplus_\sigma a_\sigma \boxtimes (M^+_{n,\lambda_\sigma} \oplus M^-_{n,\lambda_\sigma}) \ \oplus \bigoplus_\sigma b_\sigma \boxtimes M^-_{n,\lambda_\sigma} \\ \oplus \ \varrho_{31} \boxtimes M^-_{n,[31]} \ \oplus \ \varrho_{211} \boxtimes M^-_{n,[211]}. \end{aligned} \tag{16.3}$$

At $n = 4$ this has eleven nonzero joint blocks. The multiplicity spaces in $\mathscr{V}_4$, and their finite symplectic summands, are

Orthogonal input	Input dimension	Multiplicity dimension	Finite symplectic dimensions
$a_{\mathbf{1}}$	1	3876	$51 + 3825$
a_{std}	2	5440	$85 + 5355$
a_{sgn}	1	1820	$35 + 1785$
$b_{\mathbf{1}}$	2	3825	3825
b_{std}	4	5355	5355
b_{sgn}	2	1785	1785
ϱ_{31}	9	9180	9180
ϱ_{211}	9	7140	7140

Weighting the third column by the second gives 196096. Each of the eight corresponding local theta lifts is nonzero and irreducible, and they are pairwise inequivalent, by the same compact-supercuspidal argument and Howe injectivity as before. Formula (16.3) describes *particular embedded* compact-type copies in those lifts, not their full restrictions to K_n.

For a vector in an original negative ρ_{λ_σ}-isotypic space, the orthogonal isotypic projections onto a_σ and b_σ have squared-norm factors $1/3$ and $2/3$. Indeed its coset coordinate is a standard basis vector of $\mathbb{C}^3$, whose projections onto the constant line and its orthogonal complement have those norms. These are norms of vector isotypic projections, distinct from the Hilbert–Schmidt Gram factor in (15.3). In particular, a finite symplectic type in the negative summand occurs in two inequivalent local theta lifts; it does not determine the local representation uniquely.

At rank four the genuine version of every row is obtained by replacing its input ξ by $\sigma_0 \otimes \xi$. The ambient symplectic kernel still acts by minus one. This retains exactly the covering characters of Section 11.

17 First occurrence and standard modules of the higher lifts

For $\xi \in \mathrm{Irr}\, H$, let

$$\mathcal{T}_n(\xi) = \mathrm{Hom}_H(\xi, \Omega_n), \qquad r(\xi) = \min\{n \geq 0 : \mathcal{T}_n(\xi) \neq 0\}.$$

Thus $\mathcal{T}_n(a_{\mathbf{1}})$ is the former $\mathcal{T}_n$, and $\mathcal{T}_4(a_{\mathbf{1}}) = \Theta_0$. All eight inputs are self-dual: the six quotient models are real, and the two induced models are induced from real orthogonal models. Thus there is no ambiguity from replacing an input by its dual in these realizations. At rank zero, $\mathcal{T}_0(\xi) = \mathrm{Hom}_H(\xi, \mathbb{C})$.

Proposition 17.1 (Jacquet modules for irreducible orthogonal representations)**.** For $1 \le k \le n$, the normalized upper-parabolic Jacquet module is

$$R_{Q_k}\mathcal{T}_n(\xi) \simeq |\det|_2^{(k+3-2n)/2} \boxtimes \mathcal{T}_{n-k}(\xi). \tag{17.1}$$

Every nonzero $\mathcal{T}_n(\xi)$ is irreducible and equals the big and small theta lift. If $r(\xi) > 0$, its first lift is supercuspidal.

Proof. The full Weil-module Jacquet identity in Proposition 13.4 was an identity of $H \times \mathrm{GL}_k \times \mathrm{Sp}_{2(n-k)}$-modules, before taking the trivial input. Apply instead $\mathrm{Hom}_H(\xi, \cdot)$. Compact averaging on $\Omega_n \otimes \xi^\vee$ commutes with unipotent coinvariants and the modulus twist, by the identical direct- summand argument. This proves (17.1) as a full module identity. The irreducibility assertion is the supercuspidal-input theorem of [13]. At a positive first occurrence all proper maximal Jacquet modules vanish by this formula; the irreducible admissible lift is therefore supercuspidal. □

Theorem 17.2 (First-occurrence indices)**.** The first-occurrence indices and the orthogonal multiplicity dimensions $m_n(\xi) = \dim \mathrm{Hom}_H(\xi, \mathscr{V}_n)$ at ranks one and two are

ξ	$r(\xi)$	$m_1(\xi)$	$m_2(\xi)$
$a_{\mathbf{1}}$	0	5	35
a_{std}	1	1	5
a_{sgn}	2	0	1
$b_{\mathbf{1}}$	1	3	30
b_{std}	2	0	15
b_{sgn}	3	0	0
ϱ_{31}	1	3	45
ϱ_{211}	2	0	15

(17.2)

The vanishing assertion for ϱ_{211} at rank one uses the explicit finite reflection certificate in Appendix D.

Proof. The rank-one code $\mathbb{C}[D/2D^*]$ has S_3-character $\mathbf{21} + \mathrm{std}$. At rank two its code contains a free orbit of ordered independent vectors, so the sign type also occurs. The decomposition of $\mathscr{V}_n$ gives the last two columns of (17.2). These are finite orthogonal multiplicity dimensions, not entire local dimensions. Since rank zero contains only the trivial input, these nonzero entries settle first occurrence one for $a_{\mathrm{std}}, b_{\mathbf{1}}, \varrho_{31}$, and give upper bound two for $a_{\mathrm{sgn}}, b_{\mathrm{std}}, \varrho_{211}$.

Here is a direct vanishing proof for a_{sgn} at rank one. Scale a nonzero vector to a primitive $x \in M$. If two coordinates x_i, x_j are odd, choose $u_i = x_j, u_j = -x_i$, with other coordinates zero. If only x_i is odd, choose distinct $j, k \neq i$ and put $u_j = u_k = 1$, $u_i = -(x_j + x_k)/x_i$, with the remaining coordinate zero. In both cases $u \perp x$, exactly two coordinates of u are odd, and $v_2(Q(u)) = 1$. The integral reflection

$$s_u = I - \frac{2}{Q(u)} uu^T$$

fixes x and reduces to the transposition of these two odd coordinates. Thus $a_{\mathrm{sgn}}(s_u) = -1$. An equivariant Schwartz–Bruhat function must vanish at every nonzero point and also at zero. Consequently the full rank-one lift, not merely its finite code part, is zero.

Equation (15.1) implies

$$b_{\mathbf{1}} \otimes \det_H \simeq b_{\mathrm{sgn}}, \qquad b_{\mathrm{std}} \otimes \det_H \simeq b_{\mathrm{std}}.$$

The conservation relation $r(\xi) + r(\xi \otimes \det_H) = 4$ [19, Theorem A in the orthogonal–symplectic version] therefore gives $r(b_{\mathrm{sgn}}) = 3$ and $r(b_{\mathrm{std}}) = 2$. It applies over $\mathbb{Q}_2$, with rank defined as half the target symplectic dimension.

It remains to exclude ϱ_{211} at rank one. Every nonzero vector, up to scalar and H-conjugacy, has one of the eight norm square classes represented in Appendix D. For each representative x, that appendix constructs finitely many rational reflections fixing x. On ϱ_{211} their finite image has character average zero. Hence the full point stabilizer has no fixed vector. An H-equivariant map $\varrho_{211} \to \mathcal{S}(V)$, evaluated at x, gives an invariant functional for that stabilizer and is therefore zero. The same holds at every point, proving vanishing. The already displayed nonzero rank-two multiplicity completes the table. □

17.1 Higher lifts and the vanishing of Siegel Jacquet modules

For every $\xi \in \mathfrak{X} \setminus \{b_{\mathrm{sgn}}\}$, set $\tau_\xi = \mathcal{T}_2(\xi)$; these seven representations are nonzero and tempered by [15, Proposition 5.5(1)], with full source and target dimensions four and $\epsilon_0 = -1$. Put $\kappa_b = \mathcal{T}_3(b_{\mathrm{sgn}})$, which is supercuspidal by the first-occurrence theorem and Proposition 17.1. The higher-lift theorem now gives

$$\mathcal{T}_4(\xi) \simeq \begin{cases} L(\nu_2^2, \nu_2; \tau_\xi), & \xi \neq b_{\mathrm{sgn}}, \\ L(\nu_2^2; \kappa_b), & \xi = b_{\mathrm{sgn}}, \end{cases} \qquad \nu_2 = |\cdot|_2. \tag{17.3}$$

For the first line, $l = -1, r = 2$ in [15, Proposition 5.6]; for the second, $l = 4-6-1 = -3, r = 1$, so the single exponent is two. The inducing representations for $a_{\mathrm{sgn}}, b_{\mathrm{std}}, \varrho_{211}$ are supercuspidal already on Sp_4; those for $a_{\mathrm{std}}, b_{\mathbf{1}}, \varrho_{31}$ have a nonzero Klingen Jacquet module $\mathbf{1} \boxtimes \mathcal{T}_1(\xi)$, with $\mathcal{T}_1(\xi)$ supercuspidal. As before $\tau_{a_{\mathbf{1}}} = \tau_{\mathrm{ng}}$.

There is an important restriction on using the old identification method. For $\xi \in \mathfrak{X} \setminus \{a_{\mathbf{1}}\}$,

$$R_{P_{\mathbb{S}}} \mathcal{T}_4(\xi) = 0, \qquad \mathrm{Hom}_{\mathrm{Sp}_8(F)}(\mathcal{T}_4(\xi), I(s)) = 0 \quad (s \in \mathbb{C}). \tag{17.4}$$

Indeed (17.1) at $k = n = 4$ has final factor $\mathrm{Hom}_H(\xi, \mathbb{C}) = 0$, and normalized Frobenius reciprocity gives the Hom vanishing. At the function level zero evaluation is itself H-invariant, so it vanishes on every nontrivial orthogonal isotypic component. Thus the new local lifts cannot be identified by treating them as more submodules of the scalar $I(-1/2)$.

The inducing representations in (17.3) are specified by theta correspondence. Except for $\tau_{a_{\mathbf{1}}}$, their identification by independent enhanced Langlands parameters or compact-induction data remains open; see Section 18.

18 Further questions

18.1 The remaining tempered and supercuspidal inducing representations

The seven inducing representations other than $\tau_{a_{\mathbf{1}}}$ in (17.3) remain to be identified by compact-induction data or enhanced Langlands parameters. This requires identifying the lower-rank lifts and specifying the packet member and normalization. The first-occurrence indices and Jacquet modules do not alone provide that identification. In particular, $\tau_{a_{\mathrm{sgn}}}, \tau_{b_{\mathrm{std}}}, \tau_{\varrho_{211}}$ are supercuspidal on Sp_4, and κ_b is supercuspidal on Sp_6, so their proper Jacquet modules vanish.

Equation (17.4) also excludes the scalar Siegel evaluation map for all seven nontrivial orthogonal representations. Further identification requires different information, such as a compact inducing representation or a nonzero Fourier or Jacquet coefficient together with its stabilizer action.

18.2 Restrictions to maximal compact subgroups and extension groups

The space $\mathscr{V}_4$ is generated by orthogonal translates, not by the rational symplectic group. Its eleven joint summands give specific K_4-subrepresentations of the local lifts, without determining the full restrictions or their multiplicities. A description of these restrictions requires deeper lattice models and control of the intersections of their symplectic translates. The repetition of a finite K_4-type in distinct lifts is compatible with Howe duality, which distinguishes the full irreducible local representations.

The nonsplit sequences of Section 14.3 represent nonzero elements of $\mathrm{Ext}^1(X_F, A_F)$ and $\mathrm{Ext}^1(X_F, B_F)$. The dimensions of these groups in the category of smooth representations remain to be computed, as do the extension maps in the split-orthogonal Weil Jacquet filtration.

18.3 Integral models over dyadic extensions and low-rank normalizations

The local representation-theoretic classification over dyadic norm spaces does not determine the corresponding integral lattices or finite multiplicities. The two quotient actions, three-lattice geometry, and reflection computation here use the specified $\mathbb{Q}_2$-lattice; analogues over other dyadic fields require new integral calculations. The small-n Clifford representation-rank classification is likewise separate from the first-occurrence table.

For a constituent η_0 of $\tau_L|_{\widetilde{K}(2)}$ over a ramified field, its inertia subgroup and extension cocycle remain to be determined. Prescribed low-rank Paull normalizations, or changes of conductor and lattice, require separate comparisons. None of these questions is used as an assumption in the preceding proofs.

A An elementary proof of orthogonal lifting

We prove Proposition 8.2 without a discriminant-form theorem. Write $D_{r,s} = \mathrm{diag}(d_1, \dots, d_t)$, with $d_i = \pm 1$. For $g \in \mathrm{St}(T, q_{r,s})$, choose an integer lift C and set $S = C^T D_{r,s} C$. Then C is invertible over $\mathbb{Z}_2$, S_{ij} is even for $i \neq j$, and $S_{ii} \equiv d_i \pmod 4$.

The first correction $C_2 = C(I+2Y)$ must make the off-diagonal entries of $C_2^T D_{r,s} C_2$ divisible by 4 and its diagonal entries congruent to d_i modulo 8. Put $a_i = (d_i - S_{ii})/4 \pmod 2$, $c_{ij} = -S_{ij}/2 \pmod 2$. Expansion gives the equations over $\mathbb{F}_2$

$$Y_{ij} + Y_{ji} = c_{ij} \quad (i \neq j), \qquad \sum_{k \neq i} Y_{ki} = a_i. \tag{A.1}$$

They are solvable exactly when $\sum_i a_i = \sum_{i<j} c_{ij}$: after choosing each pair Y_{ij}, Y_{ji}, changing both entries changes two column sums, and these changes span the vectors of even coordinate sum. Since $g\mathbf{1} = \mathbf{1}$, each coordinate of $C\mathbf{1}$ is odd, so its square is $1 \pmod 8$. Hence

$$\sum_i a_i + \sum_{i<j} c_{ij} = \frac{\operatorname{tr} D_{r,s} - \mathbf{1}^T S \mathbf{1}}{4} \equiv 0 \pmod 2.$$

For $t = 1$ the same identity gives $a_1 = 0$.

Suppose $m \geq 2$ and $S_m = C_m^T D_{r,s} C_m$ has off-diagonal entries divisible by 2^m and diagonal entries congruent to d_i modulo 2^{m+1}. Choose a matrix X_m with entries in $\{0, 1\}$ satisfying

$$(X_m)_{ij} + (X_m)_{ji} = -\frac{(S_m)_{ij}}{2^m} \pmod 2 \quad (i \neq j), \qquad (X_m)_{ii} = -\frac{(S_m)_{ii} - d_i}{2^{m+1}} \pmod 2.$$

These conditions are independent. With $C_{m+1} = C_m(I + 2^m X_m)$, both precisions increase by one power of 2; the quadratic error is harmless because $2m \geq m+2$. The sequence converges to $C_\infty \equiv g \pmod 2$ with $C_\infty^T D_{r,s} C_\infty = D_{r,s}$. The separate initial step is what is needed in residue characteristic two.

B Principal notation and objects

Notation	Meaning
$F, \mathcal{O}, \varpi, k, q$	Dyadic local field, ring of integers, uniformizer, residue field, and its cardinality
$e, f, d = ef$	Absolute ramification index, residue degree, and $[F : \mathbb{Q}_2]$
R, A	$\mathcal{O}/2\mathcal{O}$ and $\mathcal{O}/4\mathcal{O}$; R is not a field in the ramified case
$n, m = nd$	Symplectic ranks over F and over $\mathbb{Q}_2$ after restriction of scalars
$L, K, \widetilde{K}$	$\mathcal{O}^{2n}$, $\mathrm{Sp}_{2n}(\mathcal{O})$, and its metaplectic preimage
$K(2), K(4)$	Principal congruence subgroups modulo the ideals $2\mathcal{O}, 4\mathcal{O}$
$K^\sharp(4)$	Image of the splitting of $K(4)$ acting trivially on $\mathcal{S}_L$
$\mathcal{S}_L, \tau_L$	$\mathbb{C}[R^n]$ and the finite-dimensional irreducible representation of $\widetilde{K}$
$E_{F,n}, \overline{\tau}_L$	The mod-4 double cover and its genuine Weil representation
$G_{F,n}$	The actual finite matrix image of $\overline{\tau}_L$, not necessarily equal to $E_{F,n}$
P_{full}	Faithful finite Heisenberg image, the standard dn-qubit Pauli group
J, P_J	$\varpi^{\lfloor e/2 \rfloor} R$ and the image of $\widetilde{K}(2)$
μ_{m_e}	Scalar center of $G_{F,n}$; $m_e = 4$ for odd e, and $m_e = 2$ for even e
ℓ, ι_4	Trace functional defined by the inverse different and the embedding of finite symplectic groups
$\mathrm{Mp}_m^P, \widetilde{\omega}_m^P$	Paull's mod-4 double cover and Weil representation, in symplectic rank m
AMp_m^P	The corresponding affine metaplectic μ_4-extension
$D_{r,s}, T, q_{r,s}$	Signed integral Gram matrix and its quadratic reduction in Section 8
$\Gamma_\Delta, \Gamma_O, H_{\mathrm{big}}$	Finite images and the enlarged centralizer in Section 8.4
$d_n, \mathcal{H}_{n,4}^{\pm}, P_n^{(4)}$	Dimension 2^n, invariant summands of the fourth tensor power, and the Pauli averaging projection
Π_n, C_n	The projection $P_n^{(4)}$ and its image; P_n denotes the Pauli group
N, T_N, p	The isotropic line $\langle 1111 \rangle$, quotient $N^\perp/N$, and permutation map $S_4 \to S_3$
$\Gamma_{n,4}, \widehat{\Gamma}_{n,4}, \Gamma_{O,n,4}$	Tensor image, complete symplectic image, and complete orthogonal image
$s_4, \sigma_0, \mathcal{M}(\sigma_0)$	Specified rational symplectic splitting, genuine orthogonal character, and local multiplicity representation
$D, D^*, \mathcal{A}$	Even-norm lattice, its I_4-dual, and orthogonal averaging map
$I(s), \Phi$	Normalized Siegel degenerate principal series and the evaluation-at-zero embedding $\Theta_0 \hookrightarrow I(-1/2)$
$\mathcal{T}_n, \mathcal{T}_{n,F}$	Big theta lifts of the trivial representation; n is symplectic rank, $\mathcal{T}_4 = \Theta_0$, $\mathcal{T}_0 = \mathbb{C}$
$\mathbb{D}_0, \mathbb{D}_F$	Quaternion division algebras over $\mathbb{Q}_2$ and F; scalar extension need not preserve division
$\Pi_{\mathrm{ng}}, \tau_{\mathrm{ng}}$	Specified nongeneric tempered GSp_4 summand and its irreducible restriction to Sp_4
$L(\nu^2, \nu; \tau_{\mathrm{ng}}), R_{Q_k}$	The Langlands quotient identified with Θ_0, and normalized upper-parabolic Jacquet functors
A_F, B_F, X_F	The split and anisotropic trivial-input theta submodules, and the irreducible quotient of $I_F(-1/2)$
α_H, β_H	The two distinct S_3 quotient actions, on $D/2D^*$ and on the three lattices
$a_\sigma, b_\sigma, \varrho_{31}, \varrho_{211}$	The eight irreducible representations of the compact orthogonal group
$\mathcal{A}_{\sigma,n}, \mathscr{V}_n$	Multiplicity-space average and the span of orthogonal translates
$\mathcal{T}_n(\xi), r(\xi)$	Ordinary local theta multiplicity space for input ξ, and its first-occurrence index (symplectic rank)

C Exact finite computations

The supplement contains three verification programs, their saved output, and execution instructions in `supplement/README.md`. The computations use exact arithmetic: Gaussian integer numerators over powers of two for the Weil operators, rational matrices for the lattice calculations, and finite permutations for the orthogonal actions. No floating-point tolerance is used.

Finite Weil representations. `verify_notes.py` enumerates the $\mathbb{Q}_2$ images for $n = 1, 2$ and the ramified quadratic image, including their scalar centers. It evaluates the three Paull sums on all 256 vectors, checks covariance and normalization, and enumerates the 6144 pairs defining E^{pair}. It also checks the tensor-square graph, centralizers, cubic and quartic blocks, and fourth-tensor multiplicities. For the local comparison it tests all 3840 primitive vectors modulo eight, the exact one-column compression matrices, the four-column invariant trace, and the 256 classes used in (12.6).

Normalization identities. `verify_continuation.py` checks root counts, modulus exponents, the higher-lift parameters, and transitivity of the Jacquet characters in the finite ranges recorded in `continuation_checks.json`.

Orthogonal representations and averaging. `verify_extension.py` computes the integral residue image, the order-576 permutation image on 48 points, the two quotient actions, and the eight character inner products. It checks the character of $\mathscr{V}_n$ for $1 \le n \le 4$. Hom-space tests use the distinct quotient $D^*/2D$, of order 64 per column, and verify covariance, compression and Hilbert–Schmidt norms for the examples specified in `extension_checks.json`. Appendix D gives the finite reflection computation used in the rank-one vanishing proof, including its exact character sums.

These computations check the stated finite identities. The general averaging and Jacquet identities, local irreducibility, and the cited structure and conservation theorems rely on the proofs and references in the text, not on extrapolation from finite examples.

D A finite computation for rank-one vanishing

This appendix supplies the finite step in the proof of Theorem 17.2. The reduction from all $\mathbb{Q}_2$-points to finitely many norm classes is proved first; the remaining finite character sums are computed with exact arithmetic. No assertion that the generated subgroups equal the full rational point stabilizers is required.

D.1 Eight representatives suffice

Every nonzero element of $\mathbb{Q}_2$, modulo squares, has a representative in $\{1, 3, 5, 7, 2, 6, 10, 14\}$. Indeed, remove an even power of two from its valuation and use that an odd unit is a square precisely when it is one modulo eight. For the norm $Q(x) = x^T x$, choose the vectors in the following table. They have the stated norms. If two nonzero vectors have the same norm and are different, reflection in their difference sends one to the other: anisotropy makes the denominator nonzero. Scaling a vector by a nonzero field element does not change its point stabilizer. Hence these eight vectors suffice, up to scalar and H-conjugacy, for the stabilizer-vanishing test.

For each row let

$$\mathcal{U}_x = \{u \in \{-2, -1, 0, 1, 2\}^4 : u \not\equiv 0 \pmod 2,\ x^T u = 0\}, \qquad s_u = I - \frac{2uu^T}{u^T u}.$$

These are specified rational orthogonal reflections fixing x. Let G_x be the finite permutation image of the subgroup they generate on the 48-element set

$$\mathcal{L} = \coprod_{i\in\{0,+,-\}} M_i/2M_i.$$

Here the three zero labels are distinct, because this is a disjoint union. It is not the common 64-element quotient $D^*/2D$ used in the separate Hom-average tests.

x	$Q(x)$	Distinct reflection images	$\lvert G_x\rvert$
$(1,0,0,0)$	1	10	48
$(1,1,1,0)$	3	10	48
$(1,2,0,0)$	5	10	48
$(1,1,1,2)$	7	10	48
$(1,1,0,0)$	2	10	48
$(1,1,2,0)$	6	8	48
$(1,1,2,2)$	10	7	48
$(1,2,3,0)$	14	8	48

The third column removes repetitions of the permutation image, not merely repetitions of a rational reflection matrix.

D.2 The induced character and the zero average

Use the coset representatives $h_0 = 1, h_+ = t, h_- = rt$, which also provide the lattice bases $h_iM = M_i$. The character of the actual compact-group representation $\varrho_{211} = \mathrm{Ind}_K^H \rho_{[211]}$ factors through the action on $\mathcal{L}$. Its formula is

$$\chi_{\varrho_{211}}(g) = \sum_{\substack{i\in\{0,+,-\}\\ gM_i=M_i}} \mathrm{sgn}(p_i)(\#\,\mathrm{Fix}(p_i) - 1), \quad p_i = \mathrm{red}_2(h_i^{-1}gh_i) \in S_4. \tag{D.1}$$

To justify the factorization, the 48-point action records the permutation of the cosets and, on each fixed fiber, the entire linear map on $M_i/2M_i$. Formula (D.1) is therefore the usual character formula for finite-index induction, with $\chi_{[211]}(p) = \mathrm{sgn}(p)(\#\,\mathrm{Fix}(p) - 1)$. No character table of an unidentified ambient finite group is used.

The following histograms are the output of enumerating the explicitly generated G_x. Every row in each indicated norm class has the same histogram:

$\chi_{\varrho_{211}}(g)$	-3	-1	0	1	3	9
$Q(x) \in \{1,3,5,7\}$	6	12	16	9	4	1
$Q(x) \in \{2,6,10,14\}$	4	18	16	3	6	1

The counts sum to 48 in each row, and the weighted sums are

$$-18 - 12 + 9 + 12 + 9 = 0, \qquad -12 - 18 + 3 + 18 + 9 = 0.$$

Consequently

$$\dim \varrho_{211}^{G_x} = \frac{1}{48}\sum_{g\in G_x} \chi_{\varrho_{211}}(g) = 0.$$

The full stabilizer H_x contains the subgroup generated by these reflections, so $\varrho_{211}^{H_x} = 0$ as well. Fixed vectors in the dual also vanish, by unitarity. Evaluation of an H-map $\varrho_{211} \to \mathcal{S}(V)$ at any nonzero x is such an invariant functional. At zero its value is an H-invariant functional and also vanishes. This proves $\mathcal{T}_1(\varrho_{211}) = 0$.

D.3 Verification of the finite character sums

The accompanying `verify_extension.py` enumerates the vectors in $\mathcal{U}_x$, verifies $s_u x = x$ and $s_u^T s_u = I$, computes the 48-point permutations, closes each finite group, and records both the histogram and the list of distinct reflection images' representatives in `extension_checks.json`. Its matrices use integer numerators and denominators; integrality and reduction use the 2-part of the denominator, with odd denominators units. Thus no real rounding is used to model a 2-adic congruence. Permutation closure and character averages are exact.

The enumeration verifies the two histograms. Together with the norm-class reduction, induced-character formula and stabilizer argument above, it proves the required vanishing.

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